BEV2SEG_2

Analysis of Semantic Segmentation of Planar Elements in Bird's-Eye-View and Its Applications to Automated Scene Pre-Annotation

START

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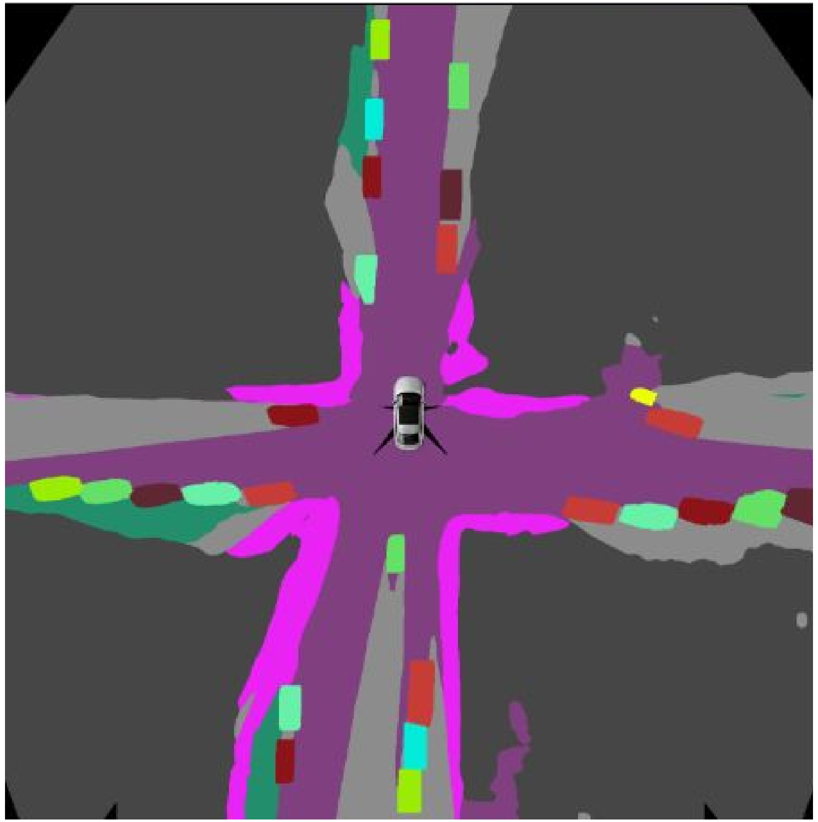
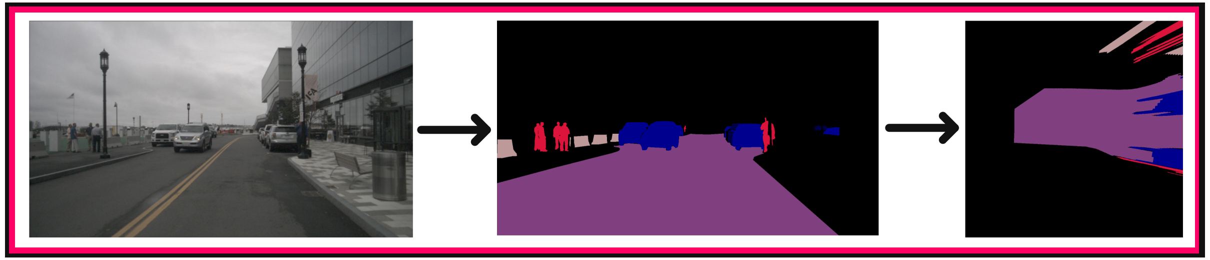
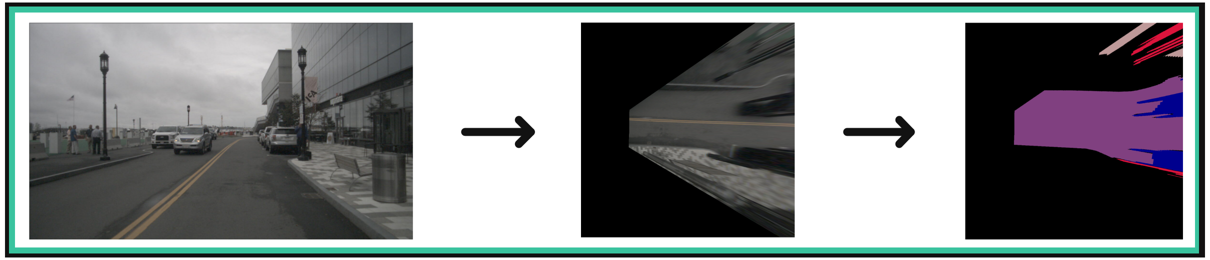
Local Semantic Bird's-Eye-View (BEV) map



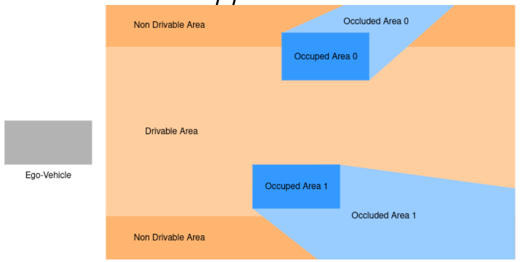
Segmenting-Then-IPM



IPM-Then-Segmenting



Pre-Annotation Application



INTRODUCTION

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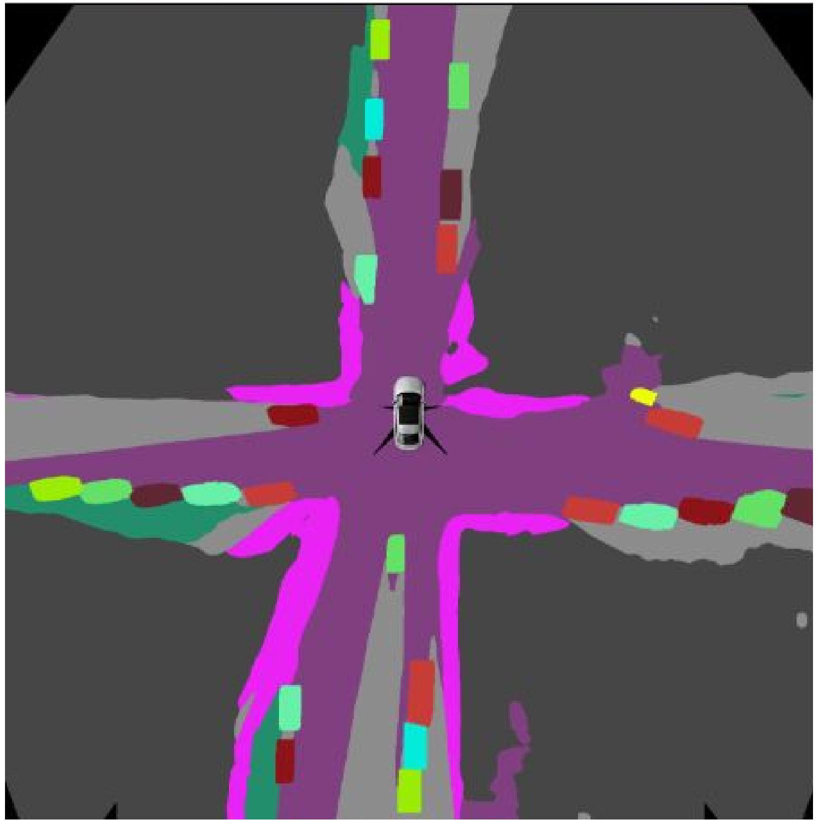

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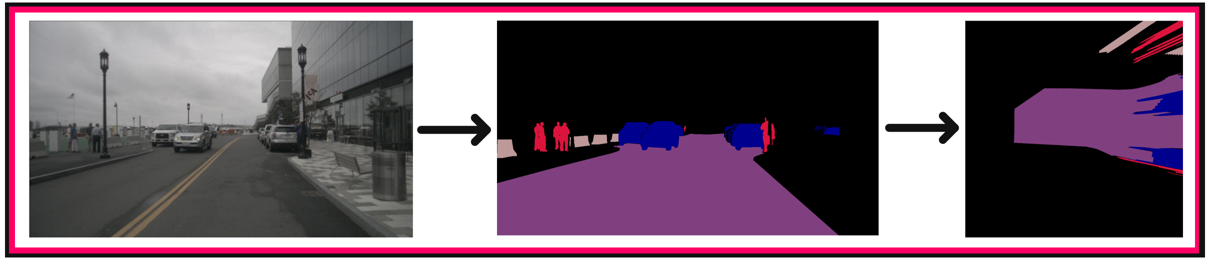

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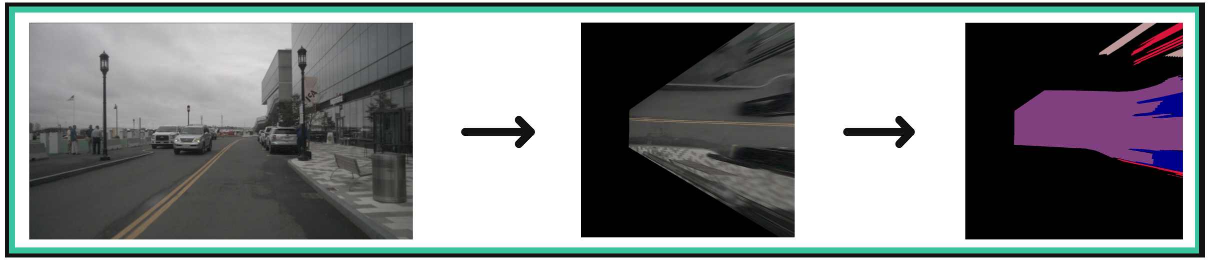

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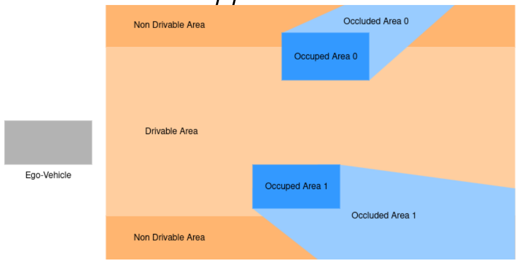

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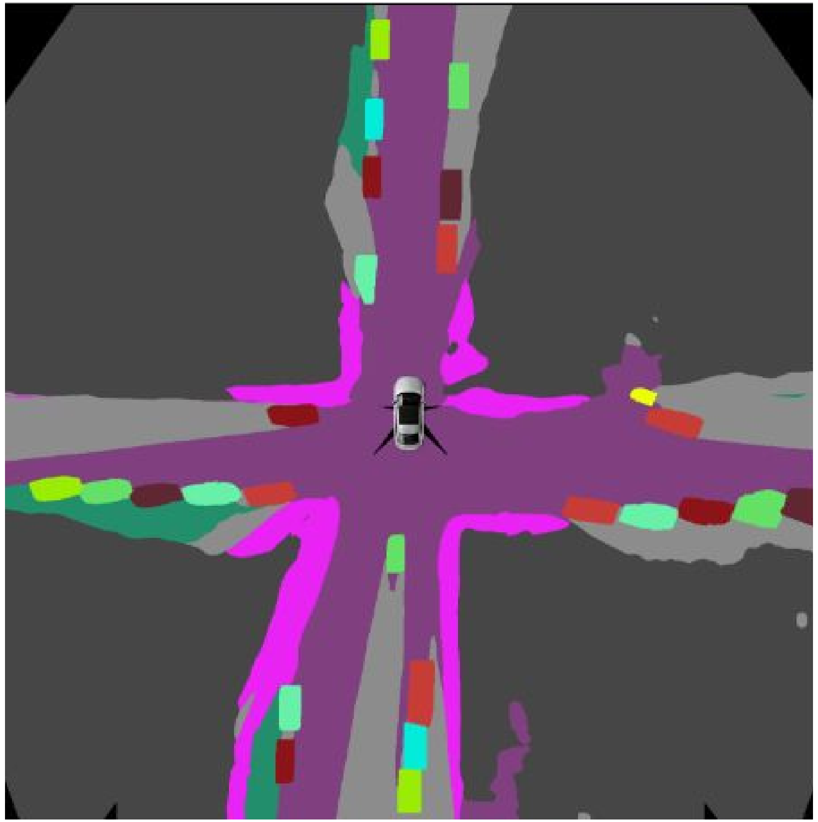
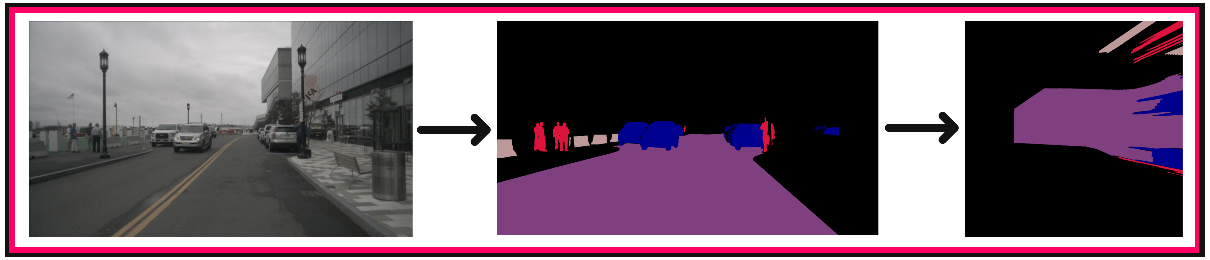
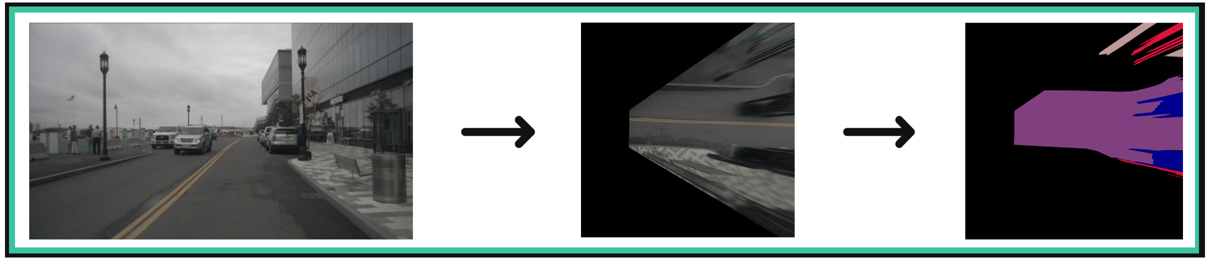


Segmenting-Then-IPM



IPM-Then-Segmenting



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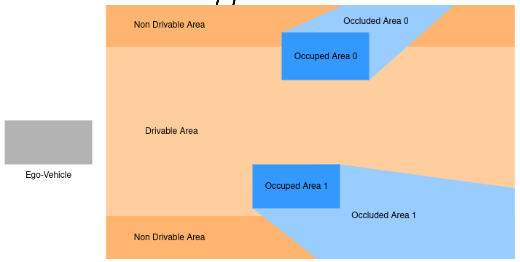

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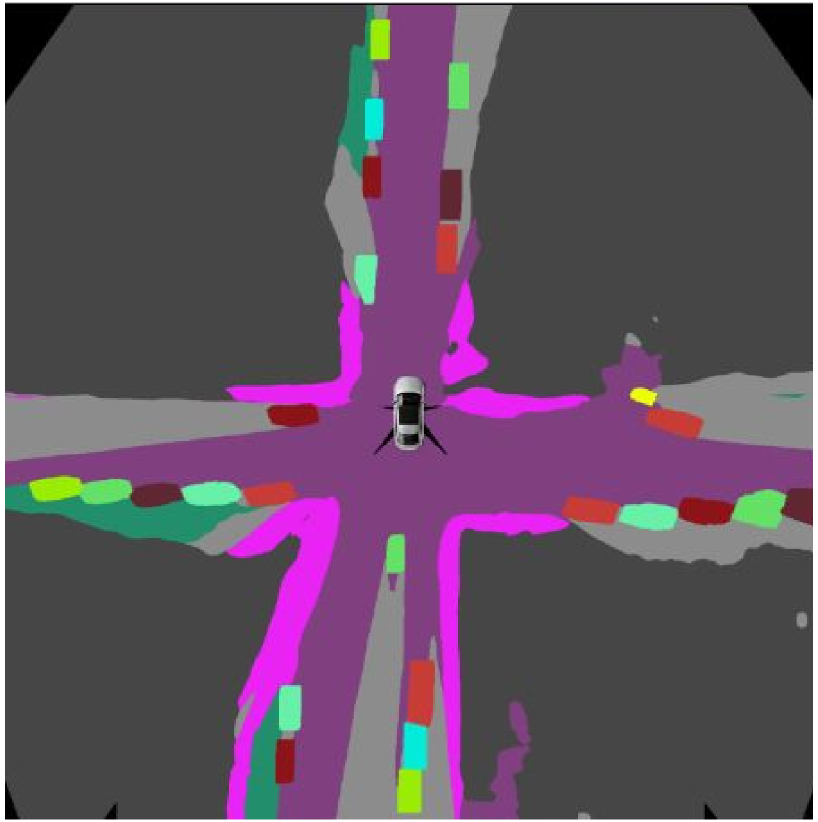

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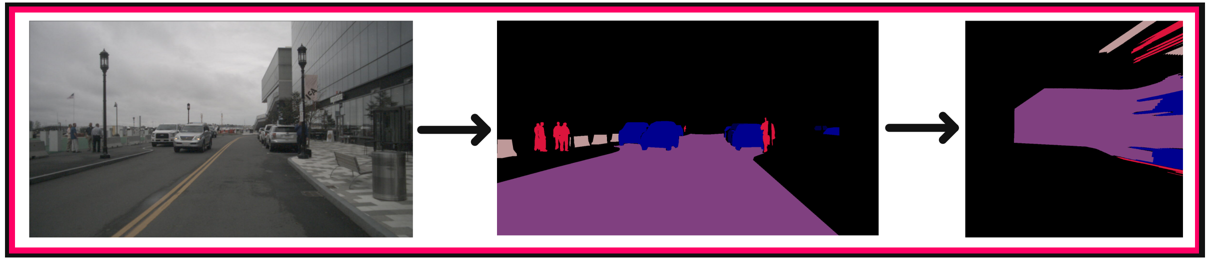

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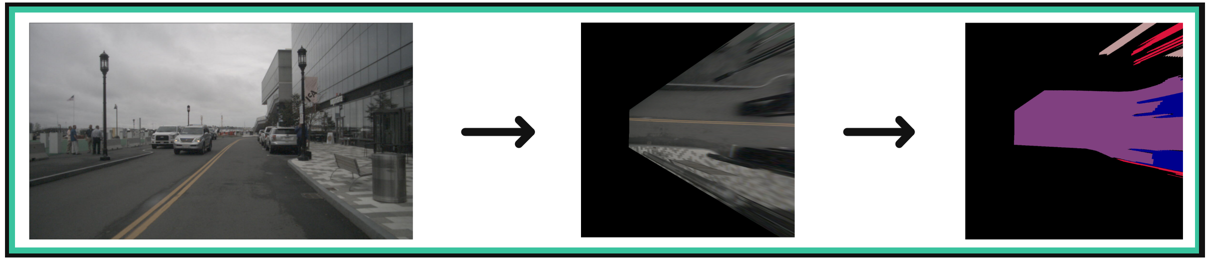

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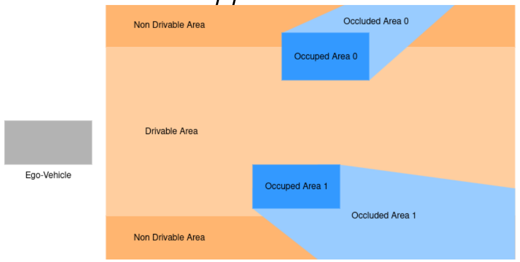


INTRODUCTION

Objectives

[HY] Does training a semantic segmentation model directly on BEV images outperforms the traditional image-space segmentation followed by IPM reprojection?

[O1] Analyze BEV semantic segmentation of plannar elements (development of BEV2Seg_2 architecture)

[O2] Develop an annotation pipeline for creating BEV occupancy/occlusion/driveable masks

[T1] Develop the BEV reprojection module

[T2] Prepare, train and evaluate the *Segmenting-Then-IPM* approach

[T3] Prepare, train and evaluate the *IPM-Then-Segmenting* approach

[T4] Perfom comparative experiments

[T5] Asses the pipeline design

[T6] Analyze pipeline strengths and limitations

[01]

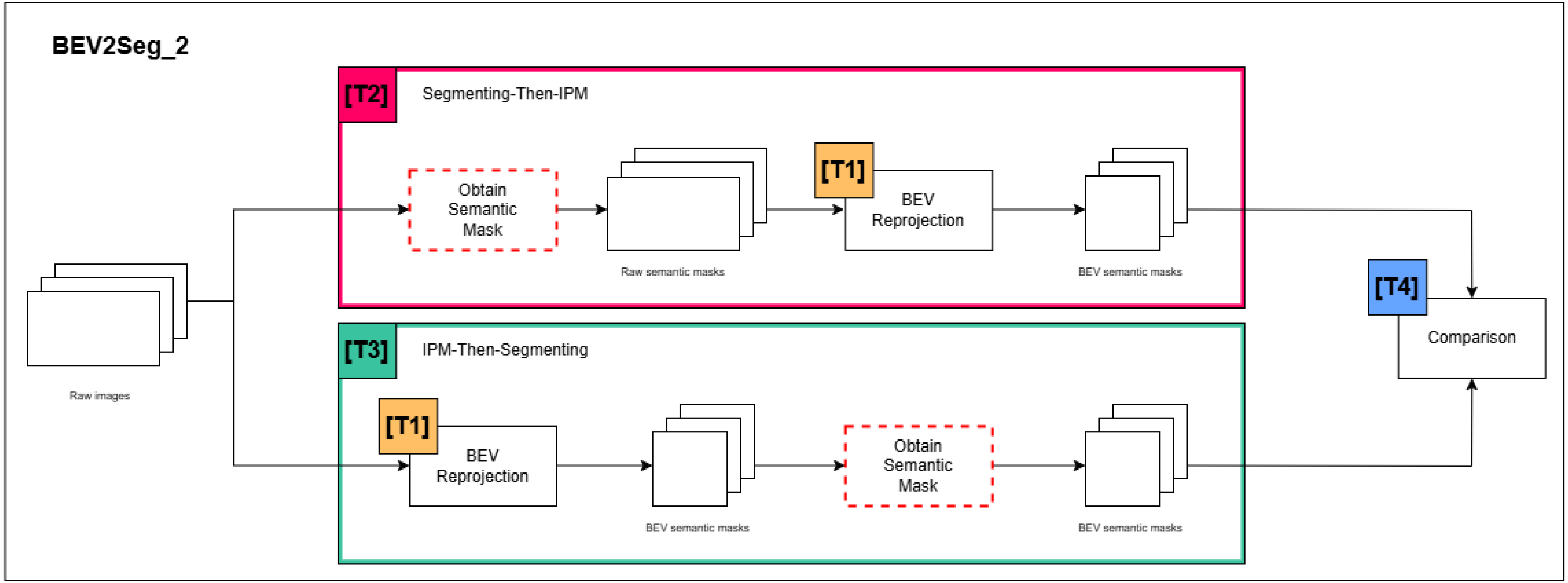
[T1] [T2] [T3] [T4]

Analyze BEV Semantic Segmentation of Plannar Elements (Development of BEV2Seg_2 Architecture)

BEV2SEG_2 FRAMEWORK

Methodology

Framework Design



BEV2SEG_2 FRAMEWORK

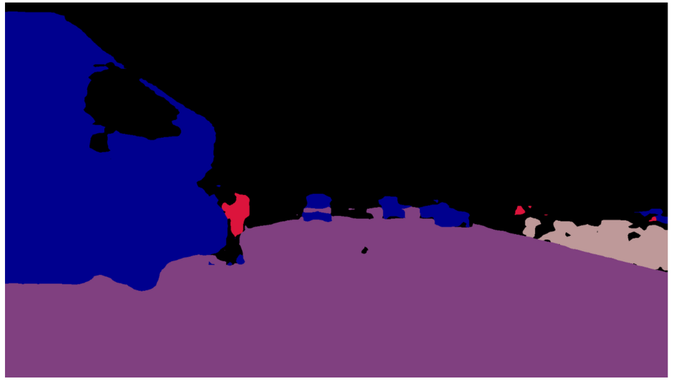
Methodology

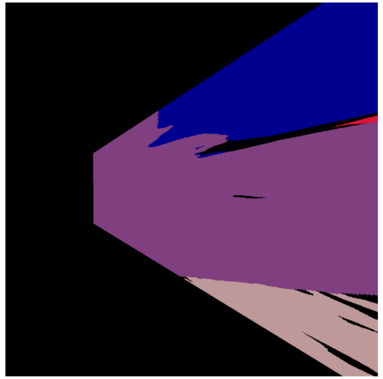
[01] [T1]

VCD
ASAM OpenLABEL®
ISO 8855:2011

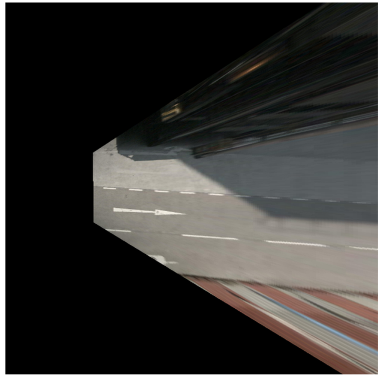
BEV Reprojection Module

30m *1024x1024*





Nearest neighbor interpolation



Linear interpolation

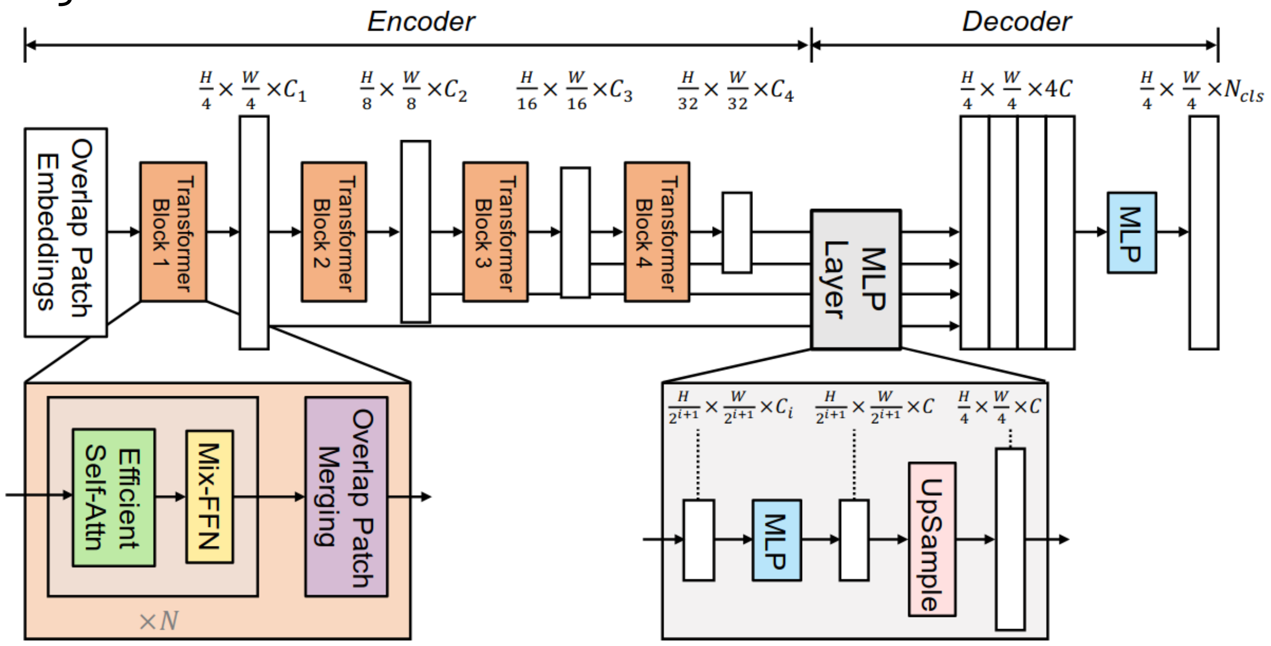
BEV2SEG_2 FRAMEWORK

[01] [T2] [T3]

Methodology

Obtain Semantic Masks ⟶ *Model Selection*

Segformer



Mix Transformer encoder (MiT)

MiT-b0, MiT-b2, and MiT-b4

Pre-trained encoders

IMAGENET

Preprocesing

SegformerImageProcessor
- Resizing 512x512
- Rescaling pixel values
- Normalizing

BEV2SEG_2 FRAMEWORK

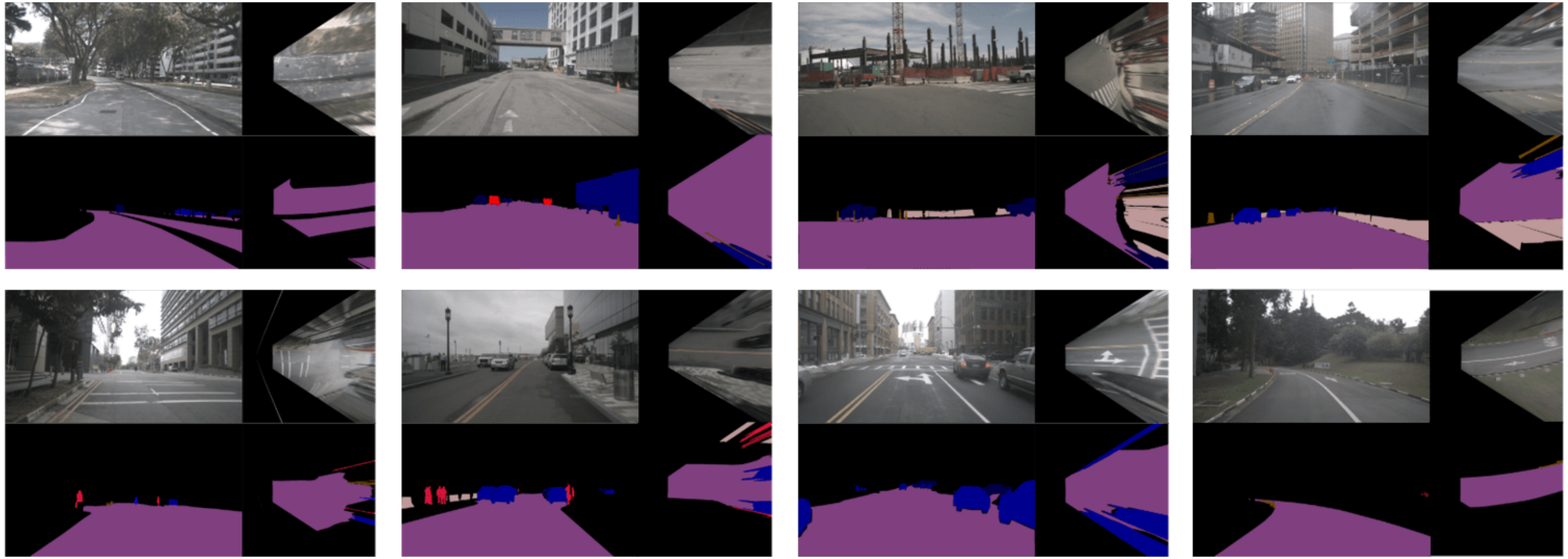
[O1] [T2] [T3]

Methodology

Obtain Semantic Masks ⟶ *Training and Evaluation Dataset*

NUSCENES by Motional NUIMAGES ⟶ OLDatasets ⟶ *BEVDataset*

- BEVDataset
  - Mini — 8 samples
    - imageToken.json
    - imageToken_bev.png
    - imageToken_bev_semantic.png
    - imageToken_raw.png
    - imageToken_raw_semantic.png
    - ⋮
  - Train — 13,178 (80%) samples
  - Validation — 1,624 (10%) samples
  - Test — 1,625 (10%) samples



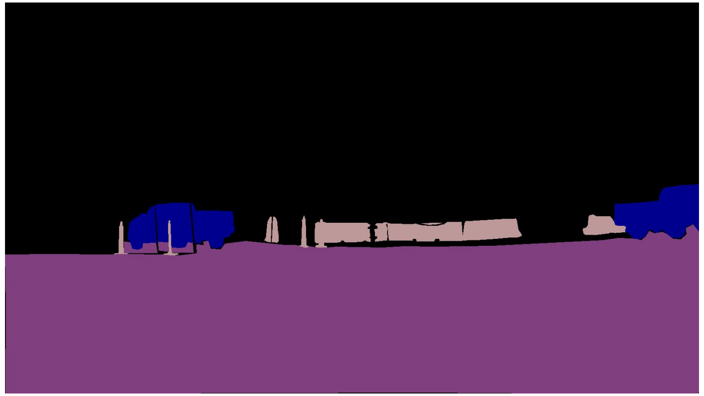
16, 427 samples; 26 semantic classes

BEV2SEG_2 FRAMEWORK

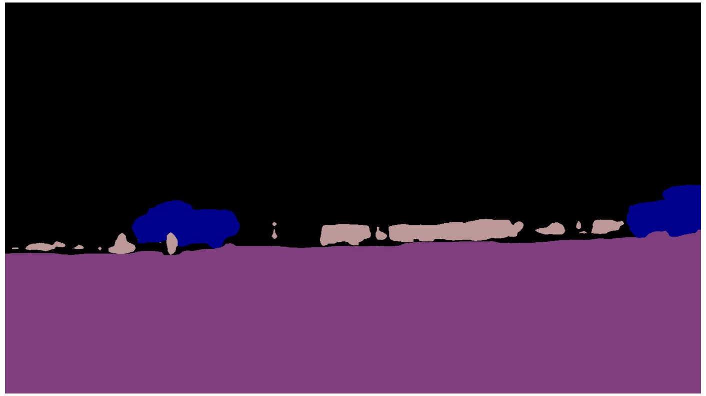
Methodology

Comparison

Original Image

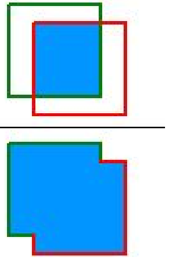


Ground-Truth Semantic Mask

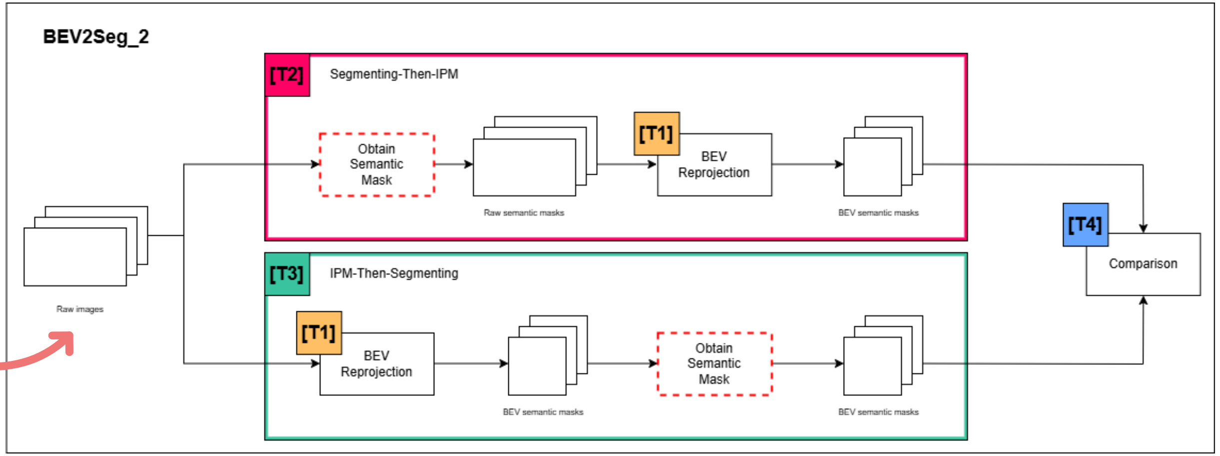


Model Predicted Semantic Mask

$$IOU = \frac{\text{area of overlap}}{\text{area of union}} =$$



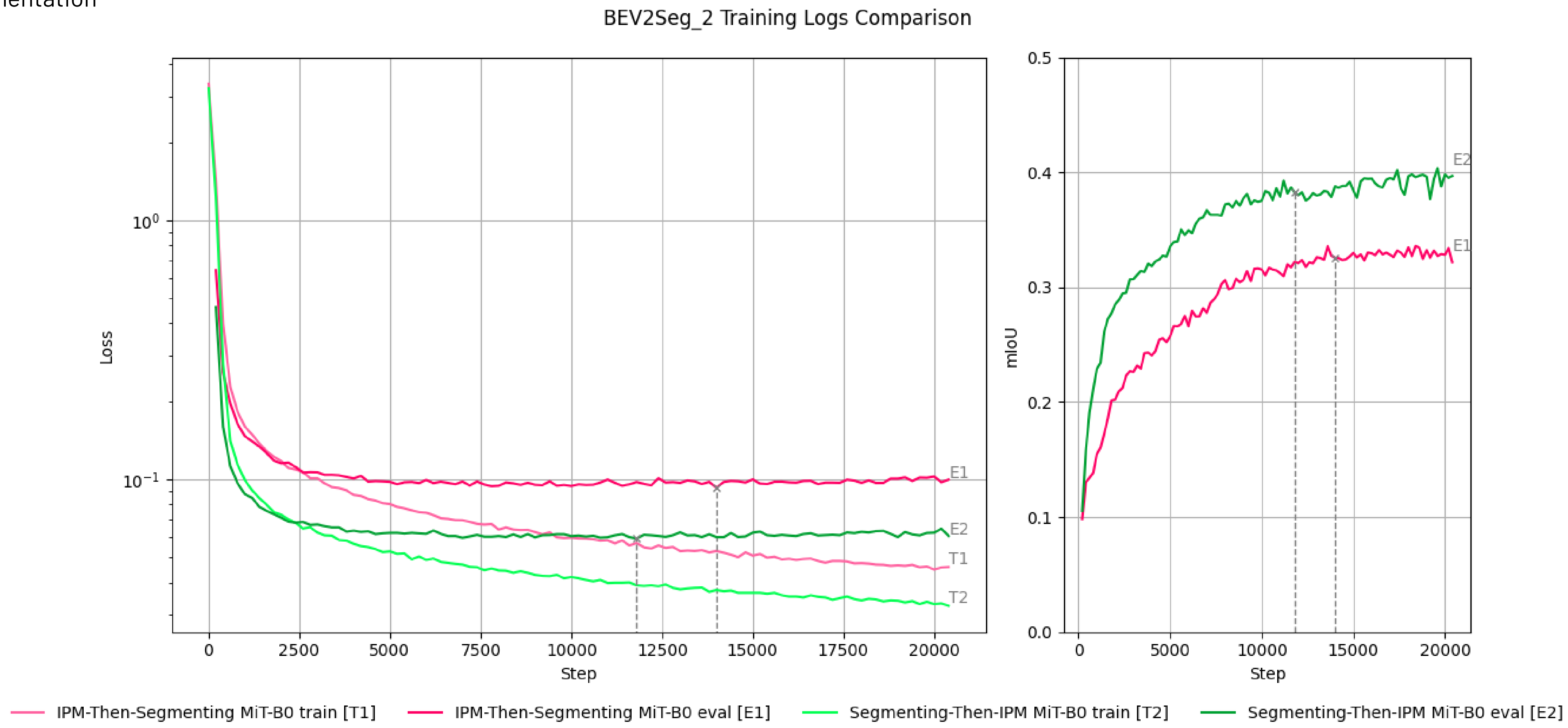
BEVDataset Test Set



BEV2SEG_2 FRAMEWORK

[01] [T4]

Experimentation

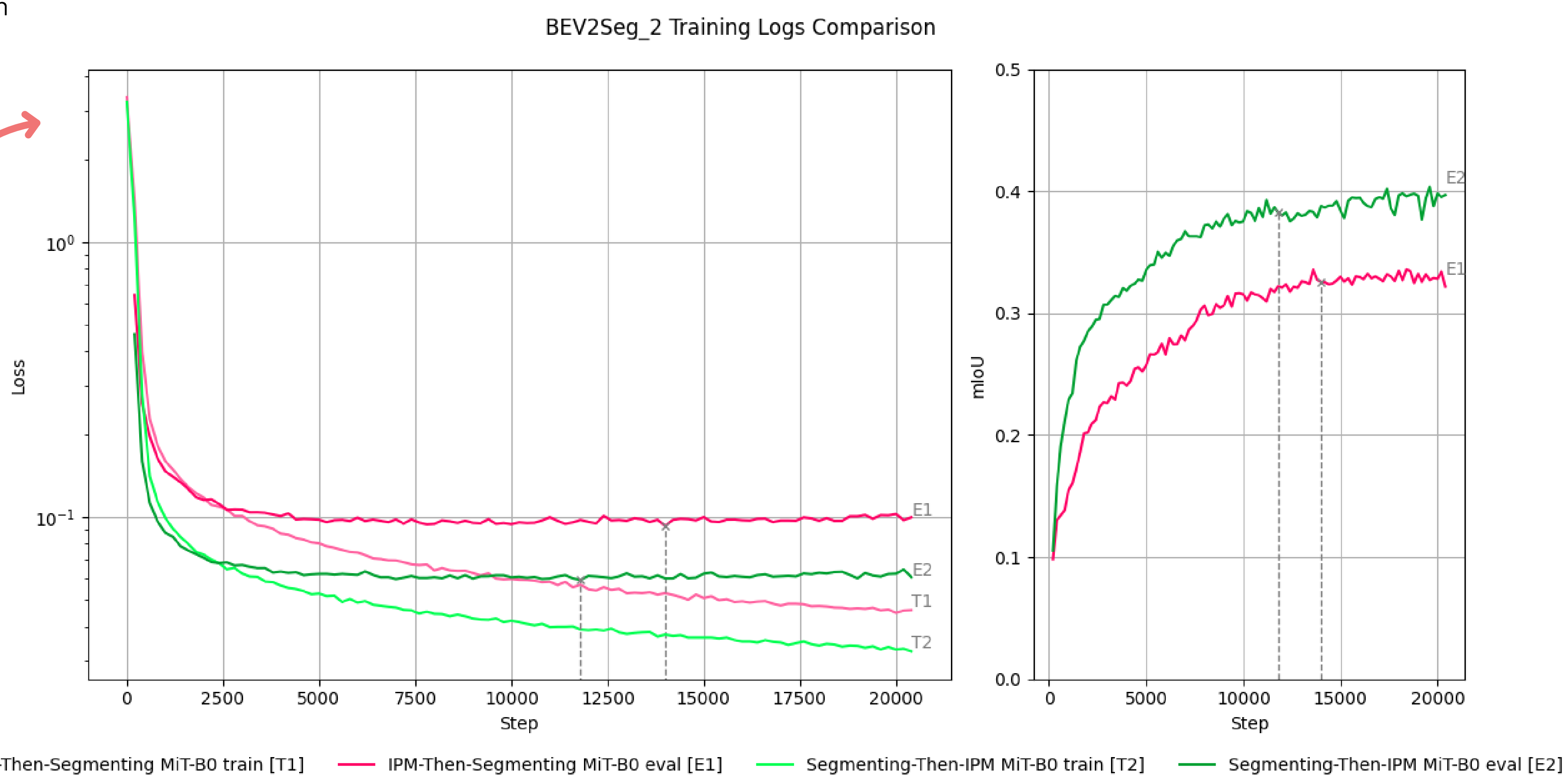


- *Overfitting*
- *Difference in metrics between the two approaches*

BEV2SEG_2 FRAMEWORK

[01] [T4]

Experimentation



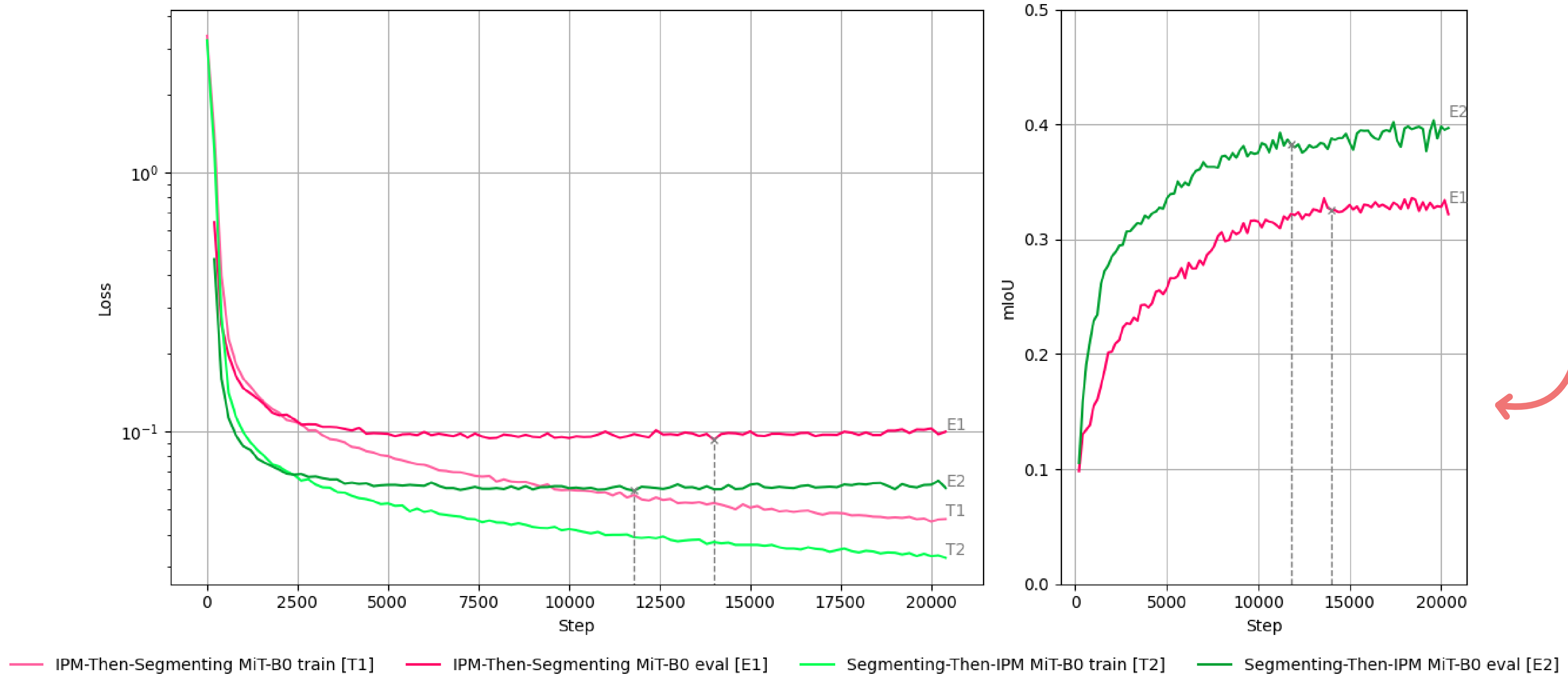
- *Overfitting*
- *Difference in metrics between the two approaches*

BEV2SEG_2 FRAMEWORK

[01] [T4]

Experimentation

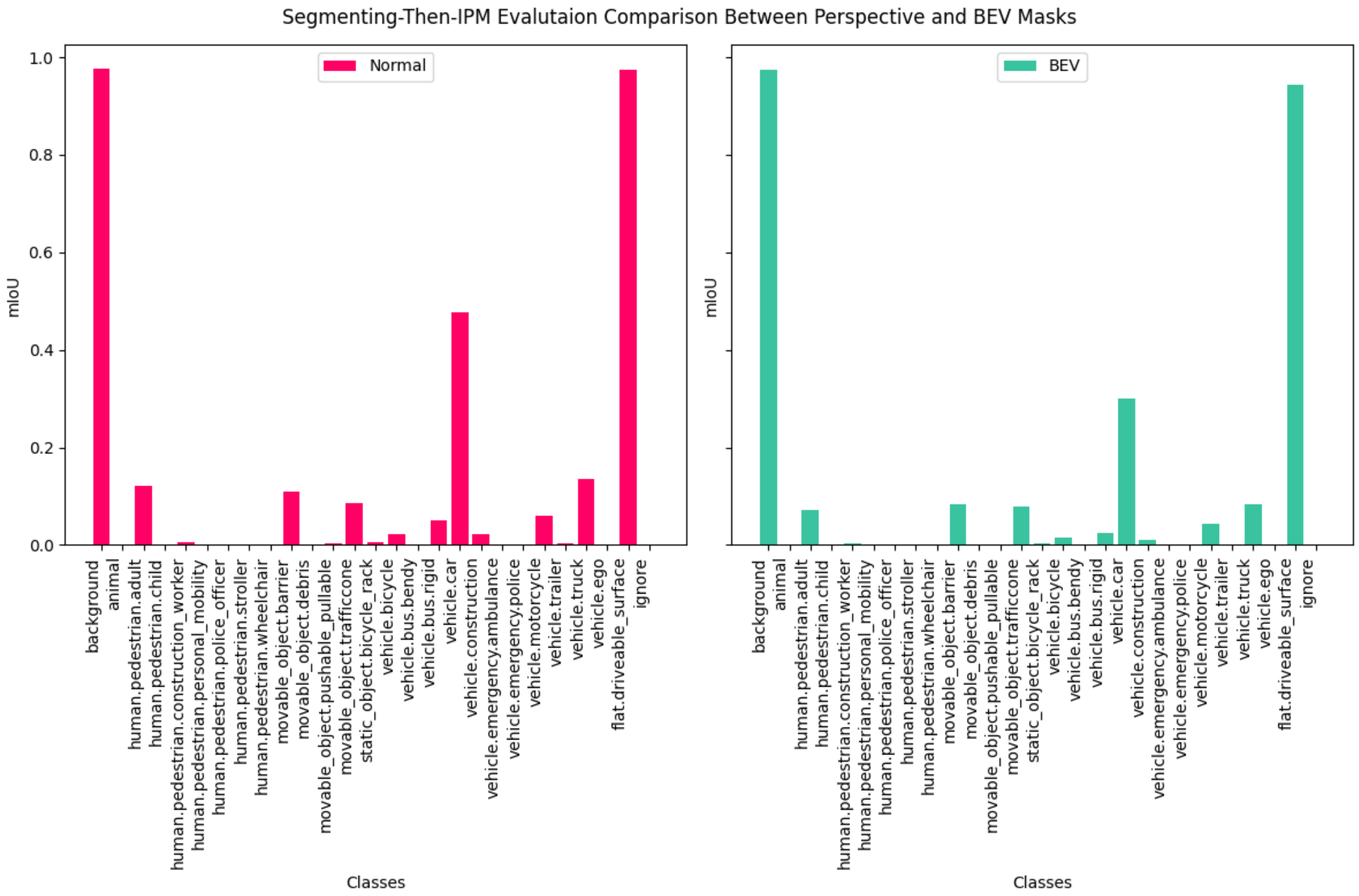
BEV2Seg_2 Training Logs Comparison



- *Overfitting*
- *Difference in metrics between the two approaches*

BEV2SEG_2 FRAMEWORK

Experimentation



- Evaluating with perspective images leads to better metric performance
- *Imbalance in performance between dominant and less frequent classes*

BEV2SEG_2 FRAMEWORK

[01] [T4]

Experimentation

- Does the label's merging strategy helps increasing the model's performance on the low presence classes?
- Which data augmentation technique is more effective for trainig a model directly on BEV images on a semantic segmentation task?
- Which of the two approaches performs better for BEV driveable area segmentation?

BEV2SEG_2 FRAMEWORK

[01] [T4]

Experimentation

Merging Labels

Merged Class	Original Label	mIoU A	mIoU B	mF1 A	mF1 B
background	background	0.97	**0.97**	0.99	**0.99**
animal	animal	0.00	0.00	0.00	0.00
human.pedestrian.adult	human.pedestrian.adult	0.07	**0.08**	0.09	**0.11**
	human.pedestrian.child	0.00	-	0.00	-
	human.pedestrian.construction_worker	0.00	-	0.00	-
	human.pedestrian.personal_mobility	0.00	-	0.00	-
	human.pedestrian.police_officer	0.00	-	0.00	-
	human.pedestrian.stroller	0.00	-	0.00	-
	human.pedestrian.wheelchair	0.00	-	0.00	-
movable_object.barrier	movable_object.barrier	0.08	**0.14**	0.09	**0.17**
	movable_object.debris	0.00	-	0.00	-
	movable_object.pushable_pullable	0.00	-	0.00	-
	movable_object.trafficcone	0.08	-	0.10	-
	static_object.bicycle_rack	0.00	-	0.00	-
vehicle.car	vehicle.bus.bendy	0.00	-	0.00	-
	vehicle.bus.rigid	0.02	-	0.03	-
	vehicle.car	0.30	**0.38**	0.33	**0.42**
	vehicle.construction	0.01	-	0.01	-
	vehicle.emergency.ambulance	0.00	-	0.00	-
	vehicle.emergency.police	0.00	-	0.00	-
	vehicle.trailer	0.00	-	0.00	-
	vehicle.truck	0.08	-	0.09	-
vehicle.ego	vehicle.ego	0.00	0.00	0.00	0.00
vehicle.motorcycle	vehicle.bicycle	0.02	-	0.02	-
	vehicle.motorcycle	0.04	**0.06**	0.05	**0.07**
flat.driveable_surface	flat.driveable_surface	0.94	**0.94**	0.97	**0.97**
		0.10	**0.32**	0.11	**0.34**

Per-class metric comparison between Model A and merged Model B evaluated with BEV images. Last row shows the mean IoU and mean F1 scores along all semantic classes of each model.

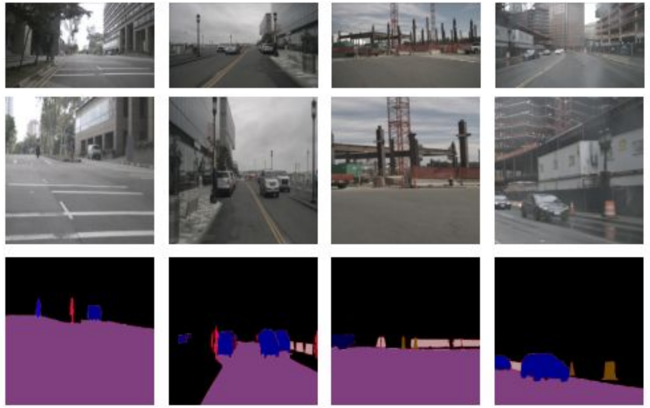
BEV2SEG_2 FRAMEWORK

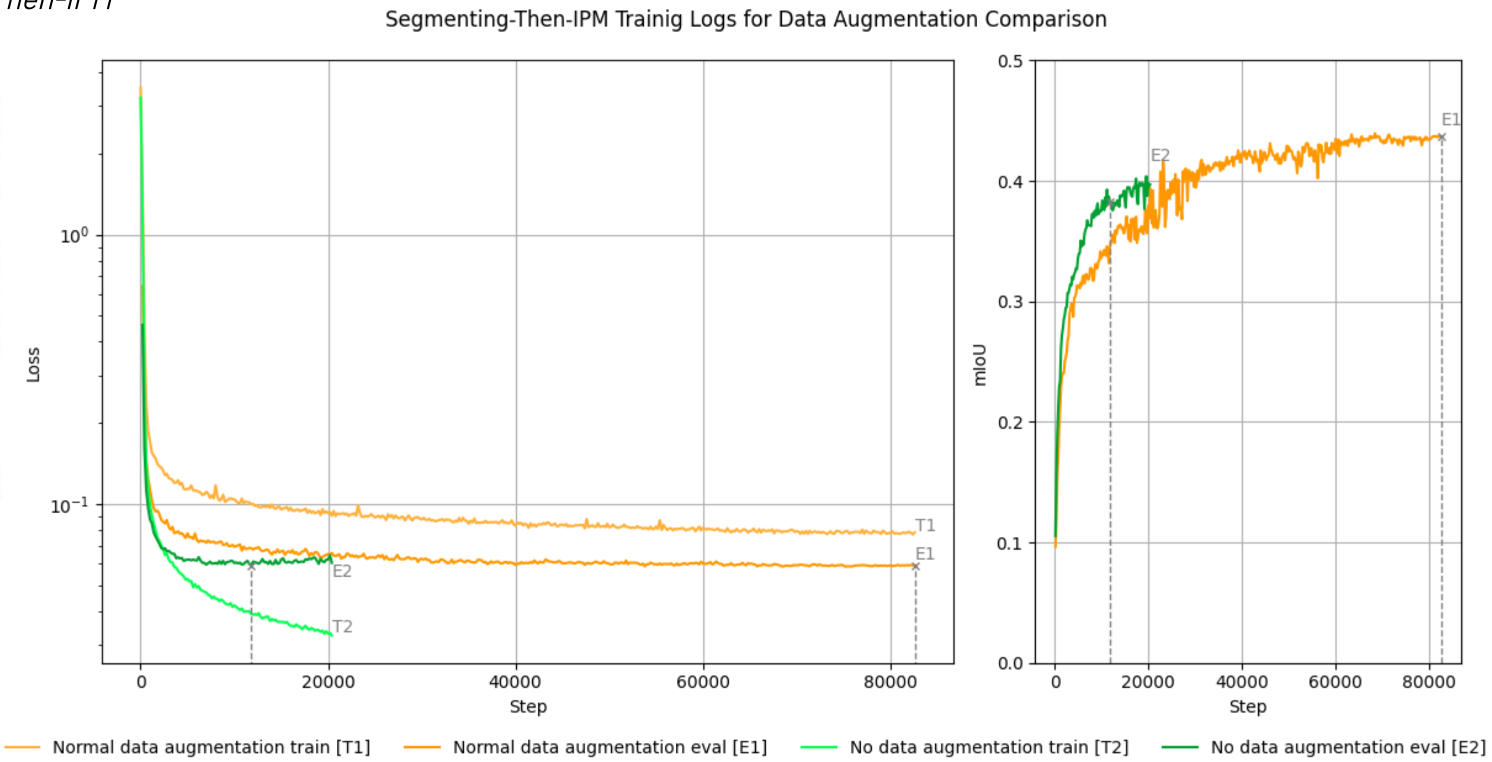
[01] [T4]

Experimentation

Data Agumentations *Segmenting-Then-IPM*

Resizing, cropping, and horizontal flipping





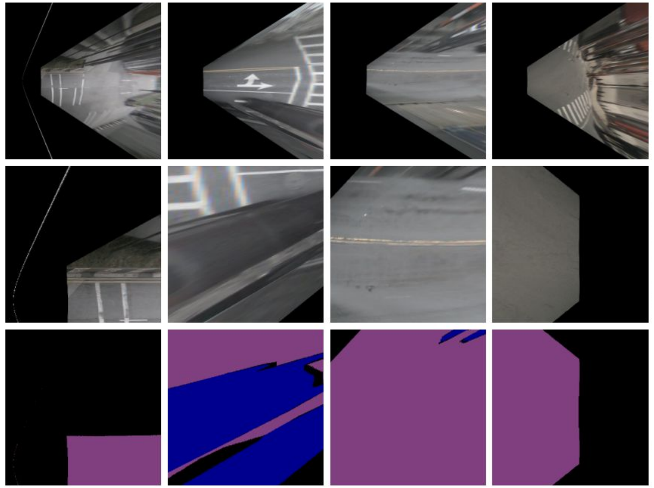
BEV2SEG_2 FRAMEWORK

[01] [T4]

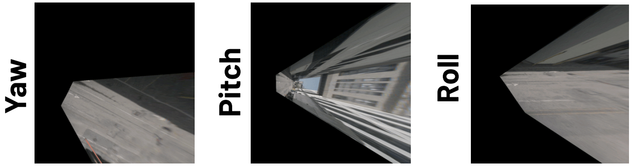
Experimentation

Data Agumentations *IPM-Then-Segmenting*

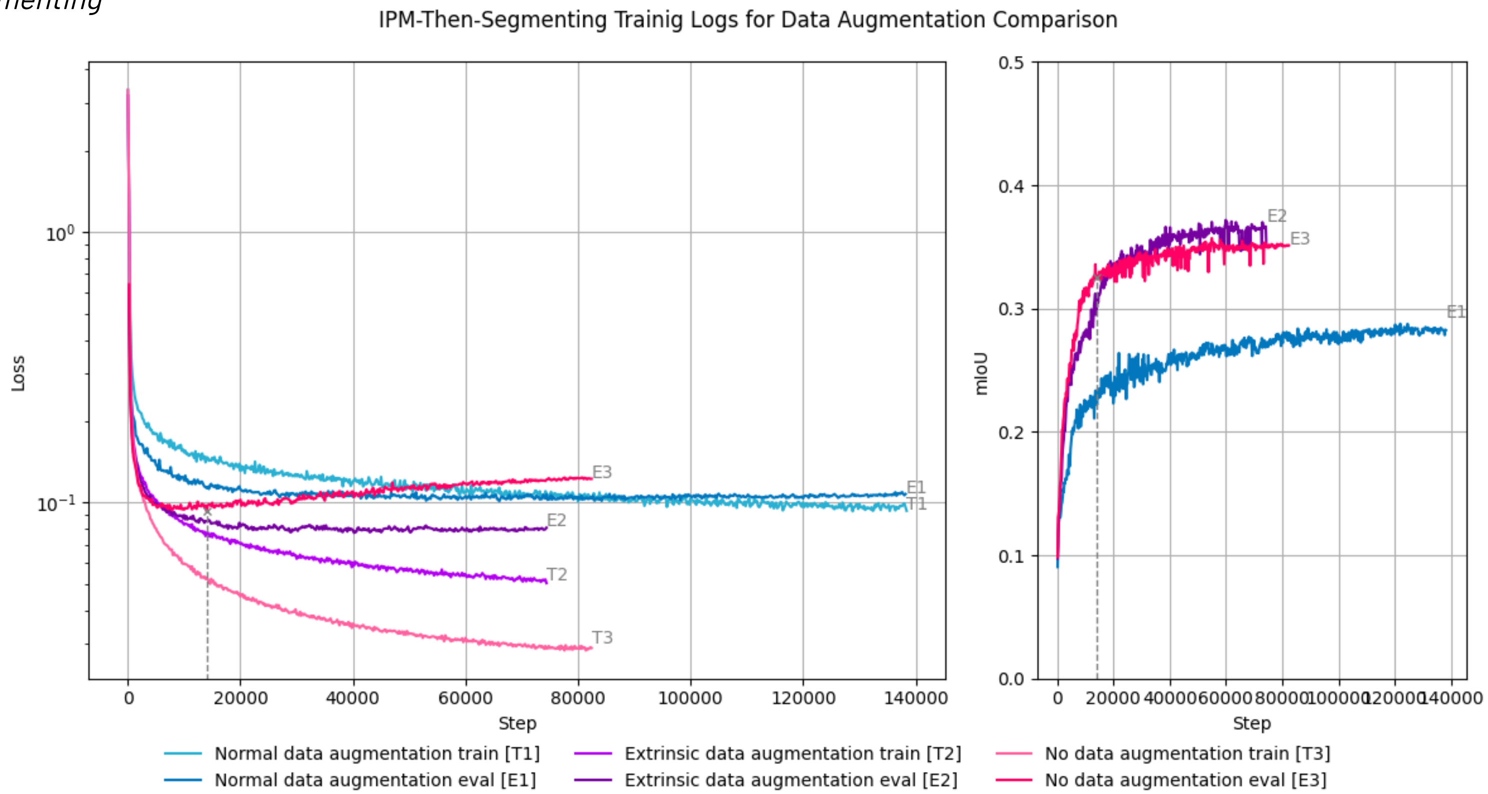
Resizing, cropping, and horizontal flipping



Camera extrinsic's parameters rotations



(-0.25rad, 0.25rad)

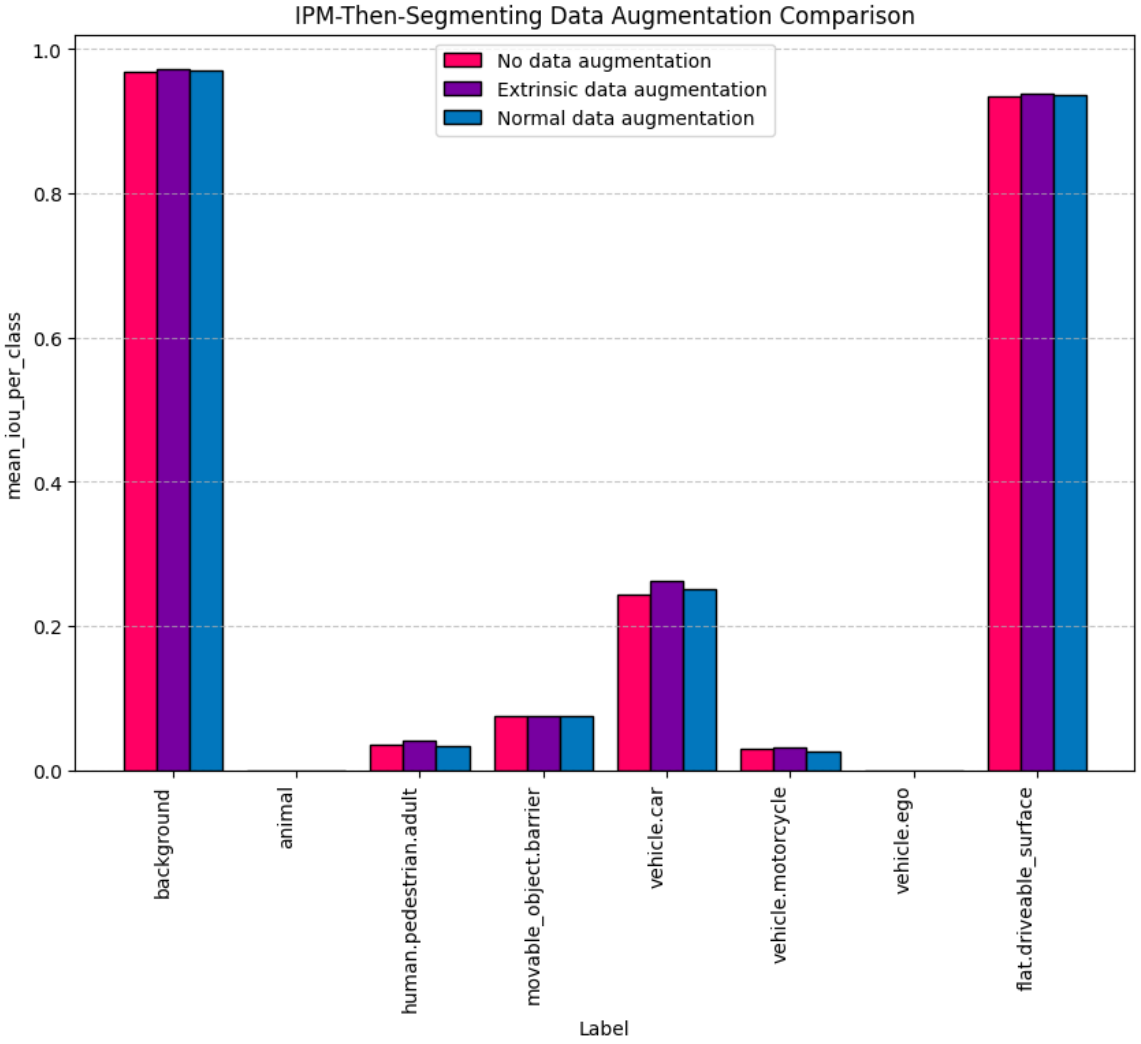


BEV2SEG_2 FRAMEWORK

[01] [T4]

Experimentation

Data Agumentations



BEV2SEG_2 FRAMEWORK

Experimentation

Segmenting-Then-IPM vs IPM-Then-Segmenting

Model	animal	background	flat. driveable _surface	human. pedestrian. adult	movable _object. barrier	vehicle. car	vehicle. ego	vehicle. motorcycle	Mean
IPM Then Segmenting									
MiT-B0 O	0.00	0.97	0.93	0.04	0.14	0.32	0.00	0.04	0.30
MiT-B0 E	0.00	0.97	0.94	0.04	0.14	0.34	0.00	0.04	0.31
MiT-B2 E	0.00	0.97	0.95	0.05	0.17	0.36	0.00	0.05	0.32
MiT-B4 E	0.00	0.98	0.95	0.06	0.17	0.36	0.00	0.06	0.32
MiT-B0 N	0.00	0.97	0.94	0.03	0.14	0.32	0.00	0.03	0.30
MiT-B2 N	0.00	0.97	0.94	0.04	0.15	0.33	0.00	0.04	0.31
MiT-B4 N	0.00	0.97	0.94	0.05	0.14	0.33	0.00	0.04	0.31
Segmenting Then IPM									
MiT-B0 O	0.00	0.97	0.94	0.08	0.14	0.38	0.00	0.06	0.32
MiT-B0 N	0.00	0.98	0.95	0.09	0.18	0.45	0.00	0.06	0.34
MiT-B2 N	0.00	0.98	**0.95**	**0.11**	**0.20**	**0.48**	0.00	0.08	**0.35**
MiT-B4 N	0.00	**0.98**	0.95	0.11	0.20	0.47	0.00	**0.08**	0.35

O: Overfitted. N: Normal. E: Extrinsic.

Comparison of mIoU scores across models. Bold values indicate the best- performing model for each semantic class.

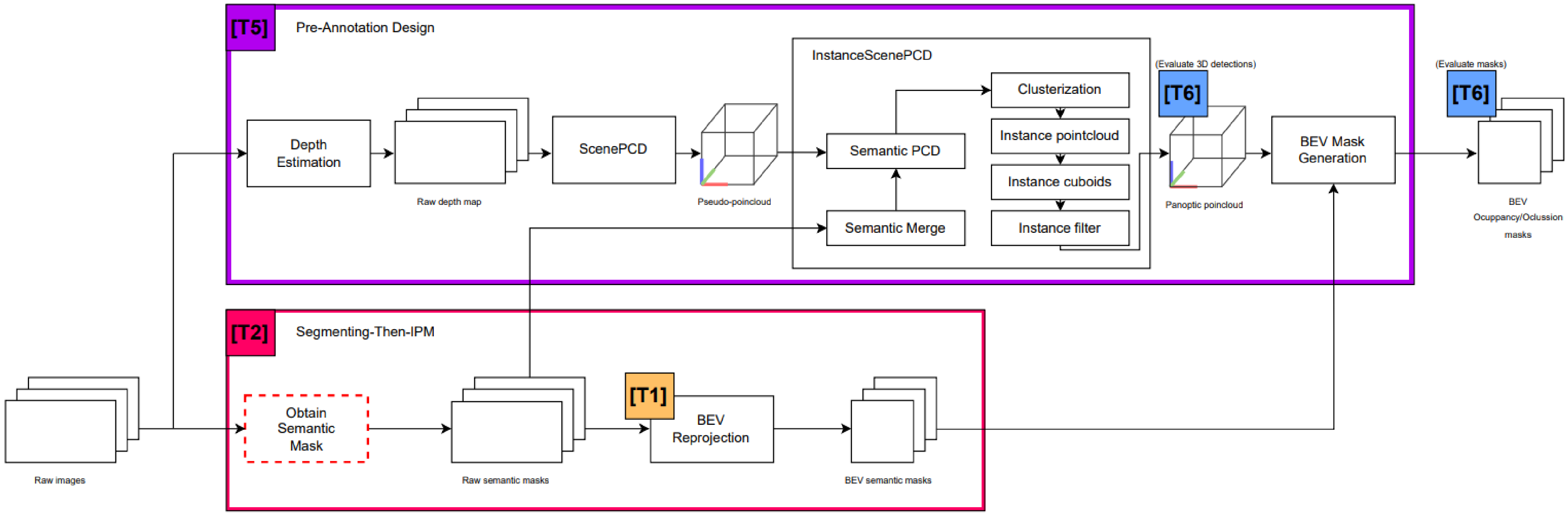
[02]

Develop an Annotation Pipeline for Creating BEV Occupancy/Occlusion/Driveable masks

[T5] [T6]

PRE-ANNOTATION PIPELINE

Pipeline's Design

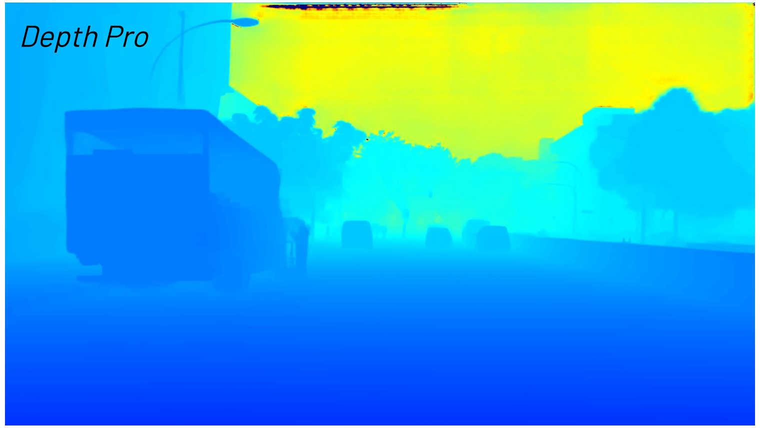


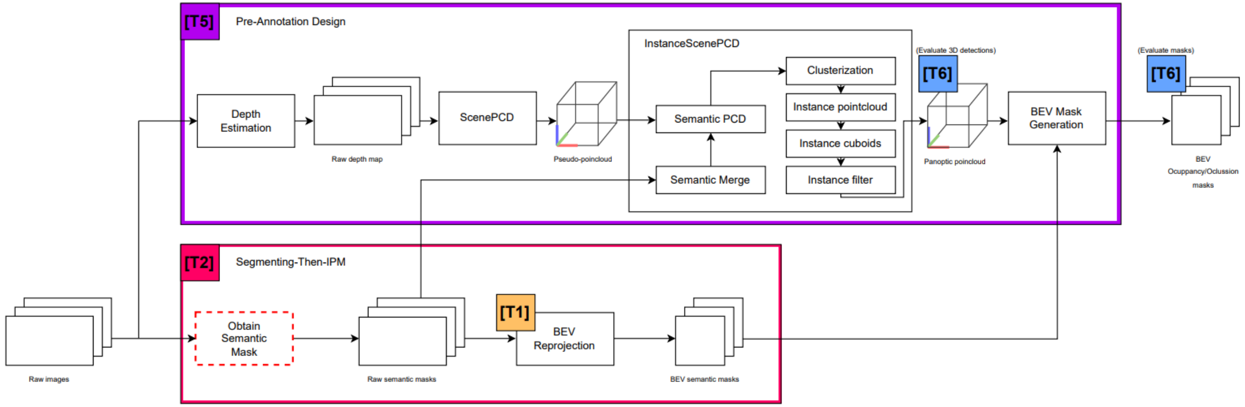
PRE-ANNOTATION PIPELINE

Pipeline's Design

Input Image

Depth Estimation



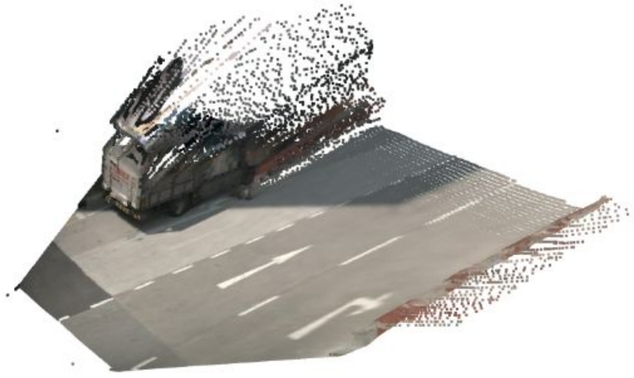


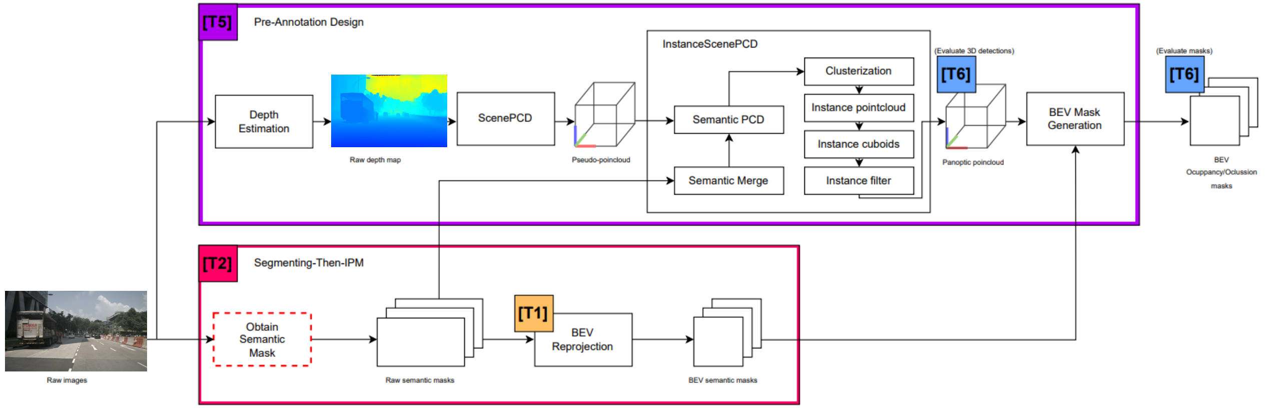
PRE-ANNOTATION PIPELINE

[02] [T5]

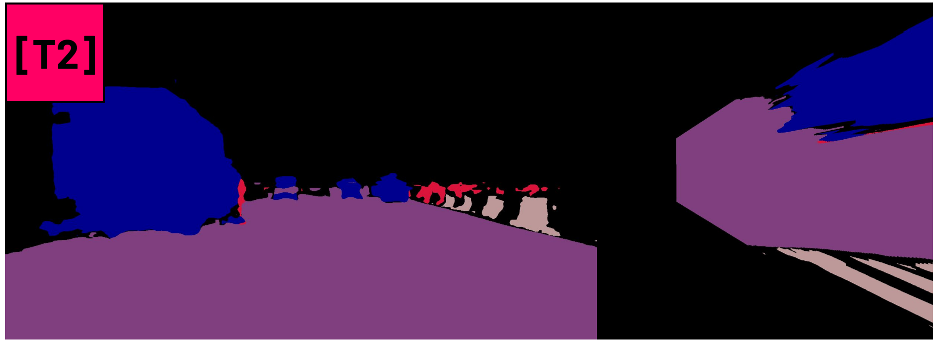
Pipeline's Design

ScenePCD





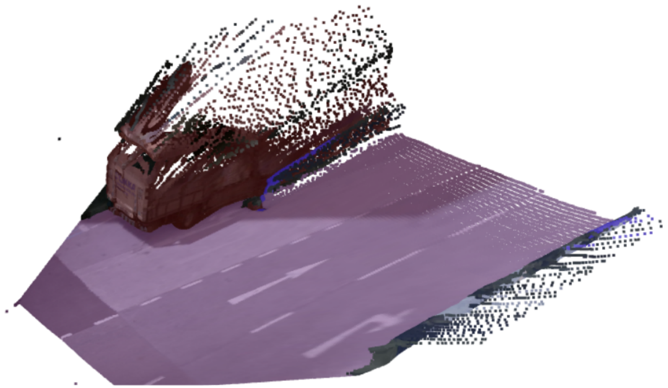
Semantic Mask Generation

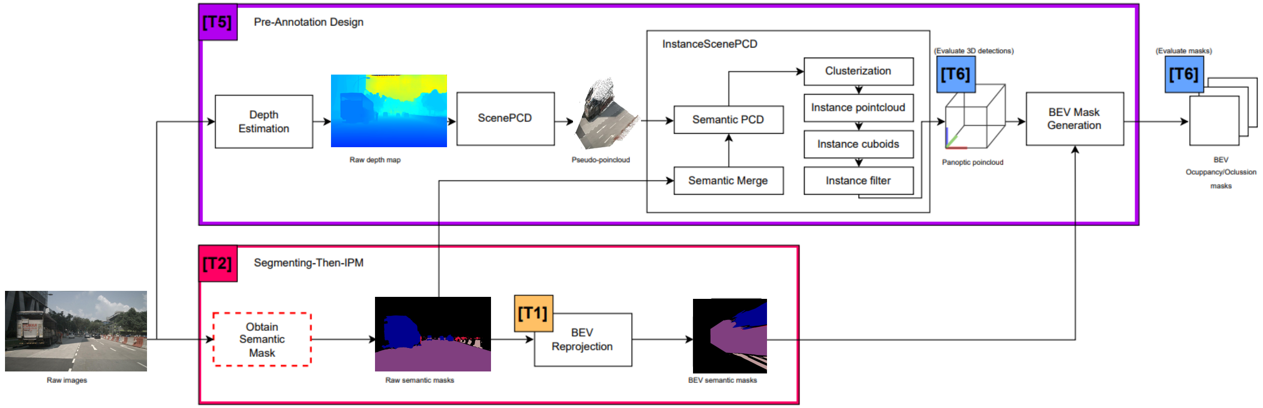


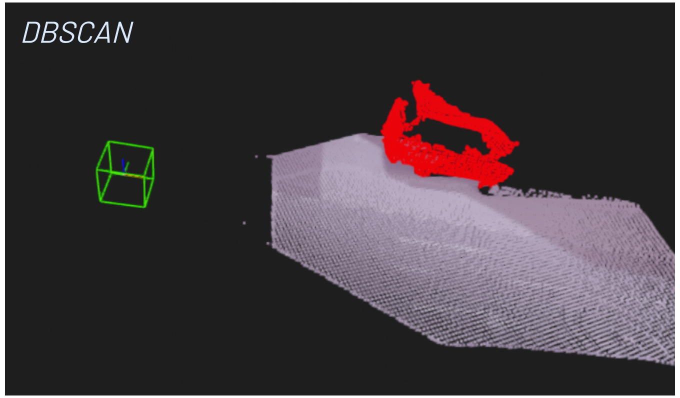
PRE-ANNOTATION PIPELINE

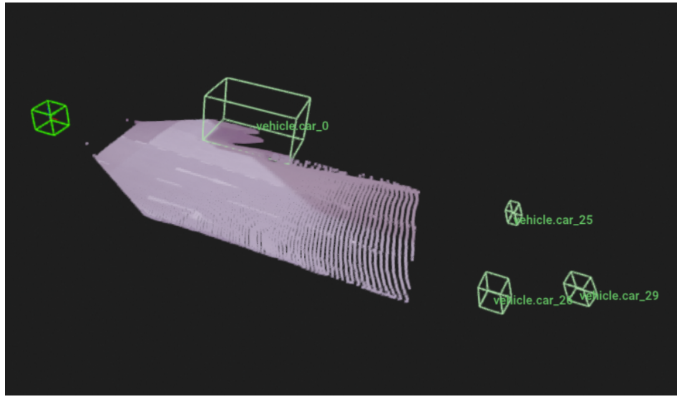
Pipeline's Design

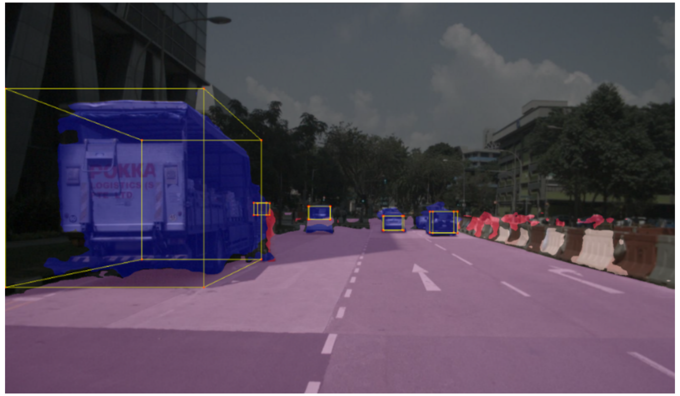
InstanceScenePCD







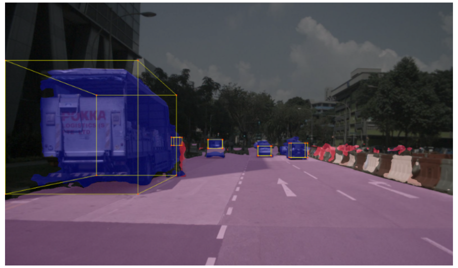




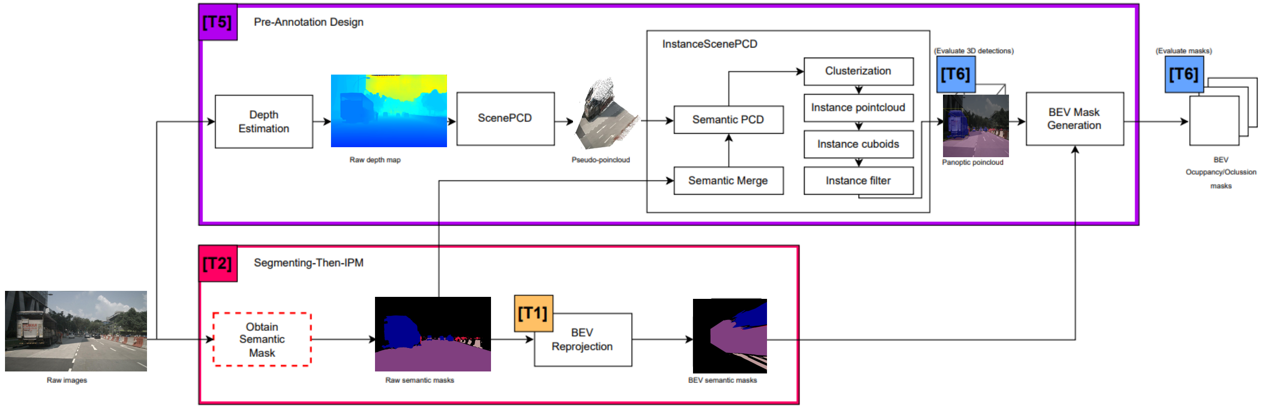
PRE-ANNOTATION PIPELINE

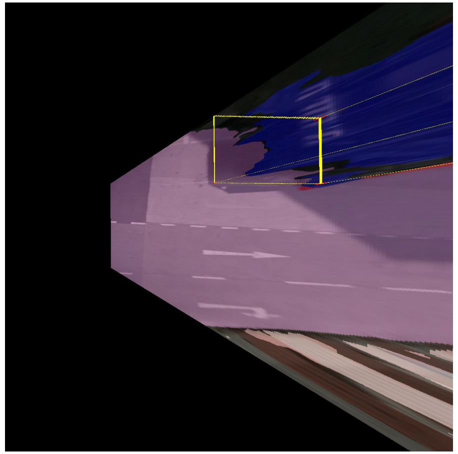
Pipeline's Design

BEV Mask Generation

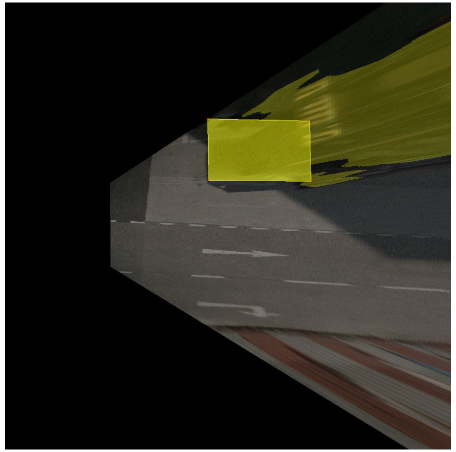


Panoptic Pointcloud

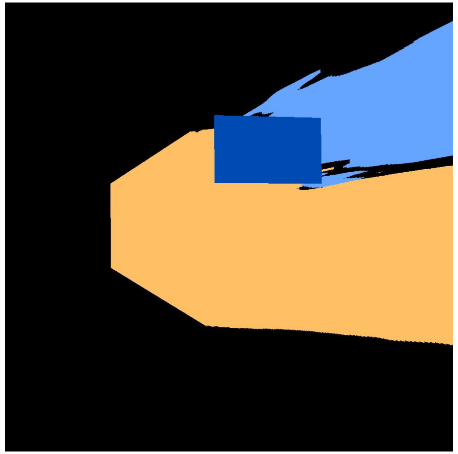




Cuboid Projection in BEV



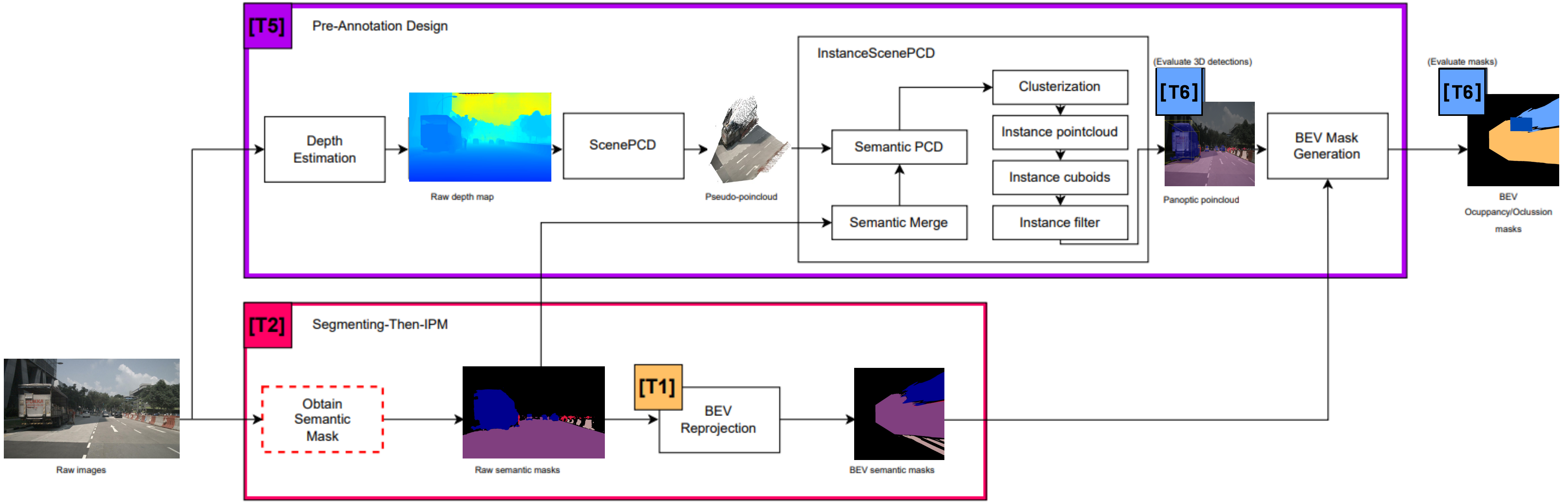
Occupancy/Occlusion Association



Final Occupancy/Occlusion/Drivable Mask

PRE-ANNOTATION PIPELINE

Pipeline's Evaluation



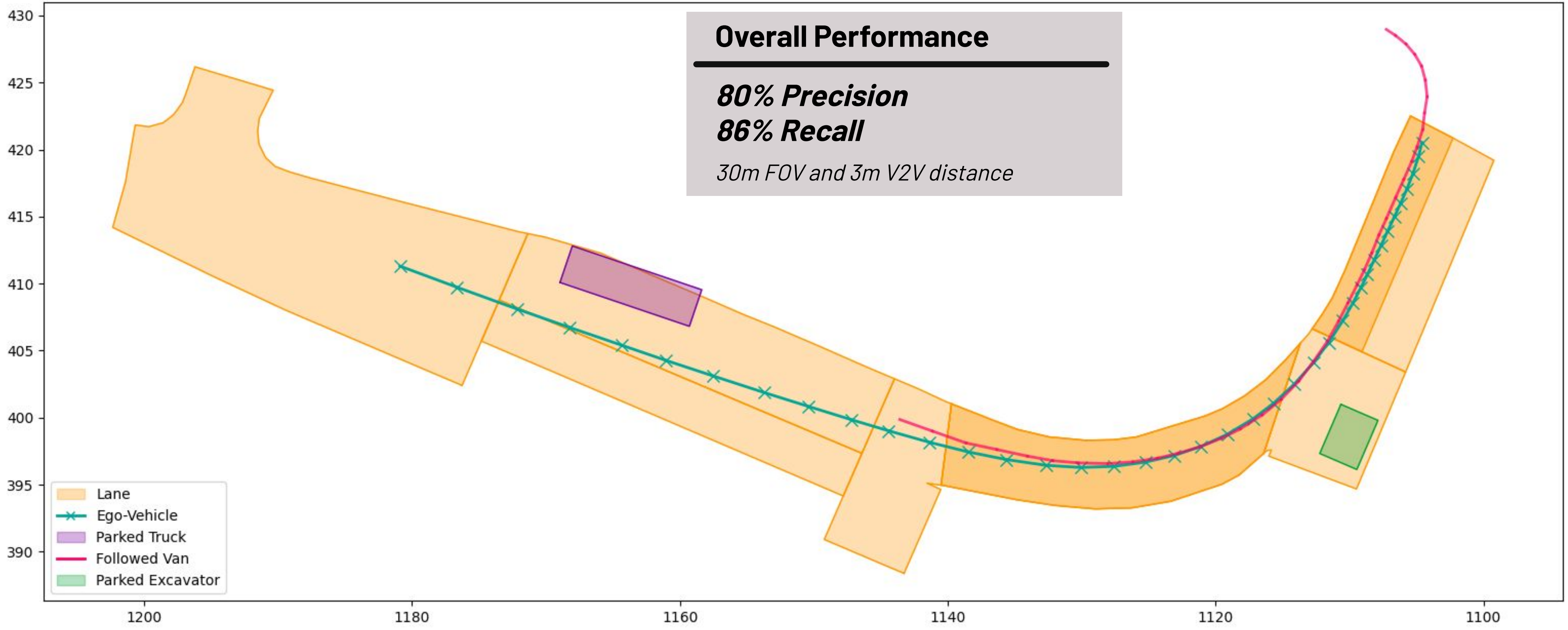
[T6]

- 3D Detections Evaluation
- BEV Masks Evaluation

PRE-ANNOTATION PIPELINE

Pipeline's Evaluation

3D Detections Evaluation

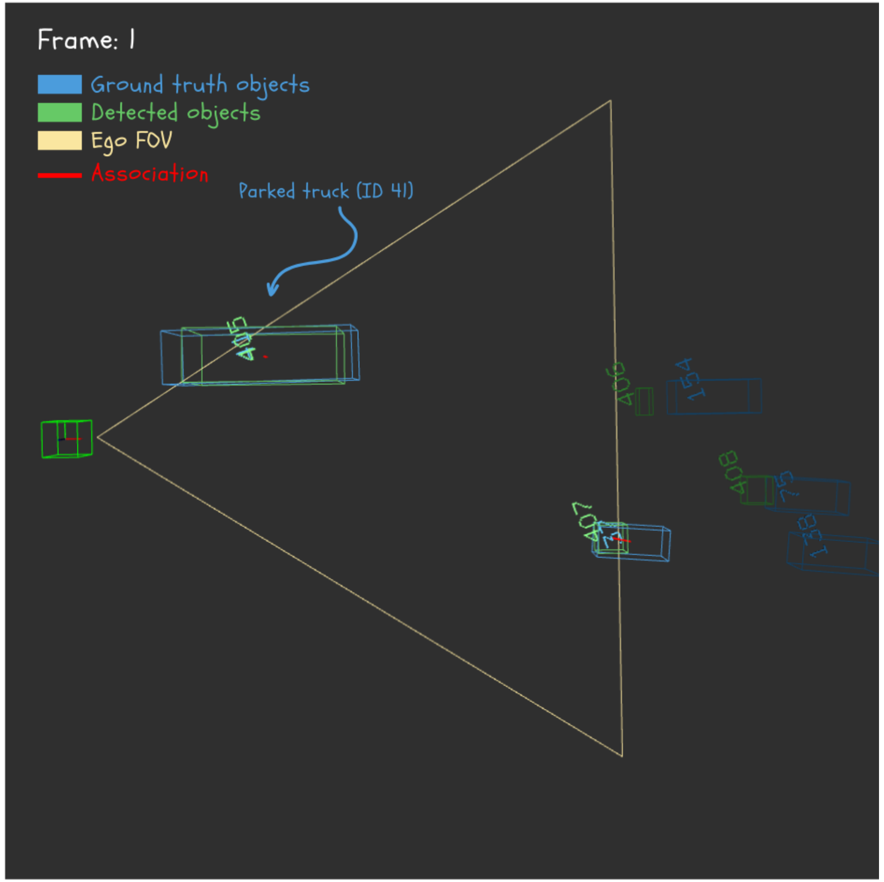


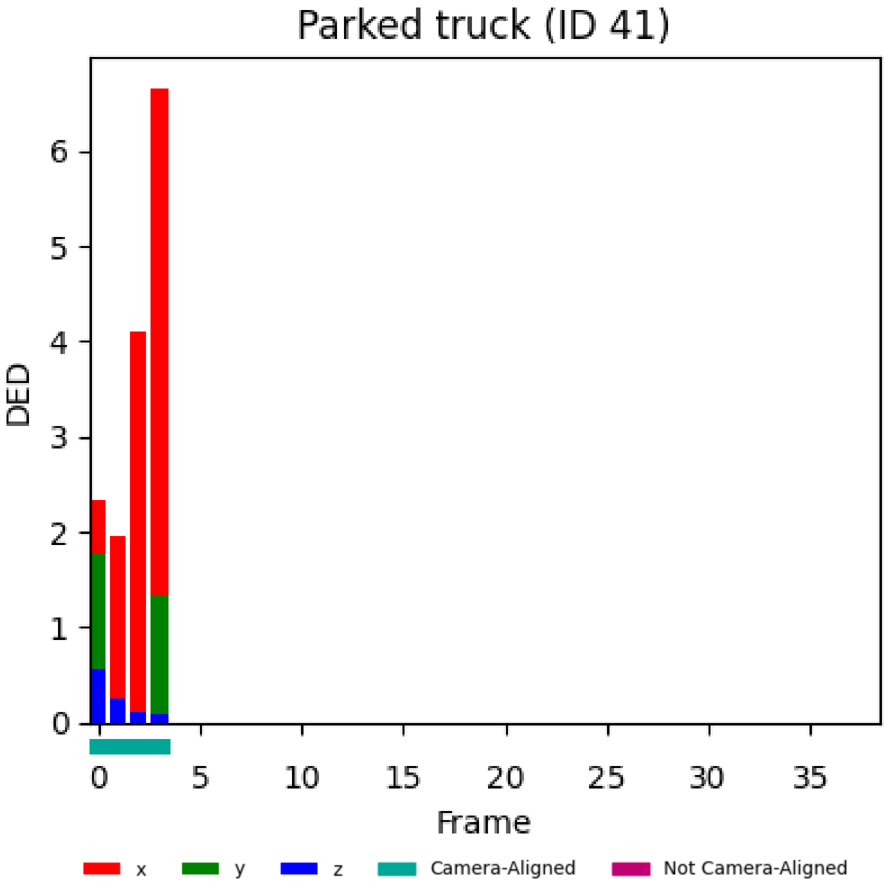
PRE-ANNOTATION PIPELINE

Pipeline's Evaluation

3D Detections Evaluation

Parked Truck





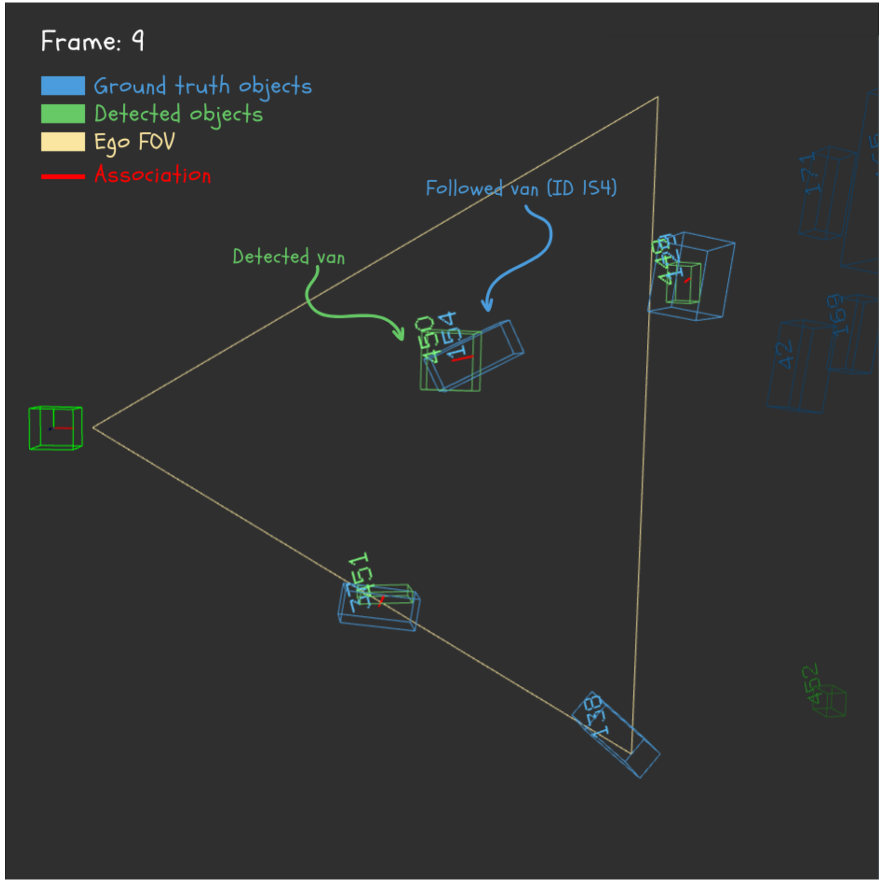
PRE-ANNOTATION PIPELINE

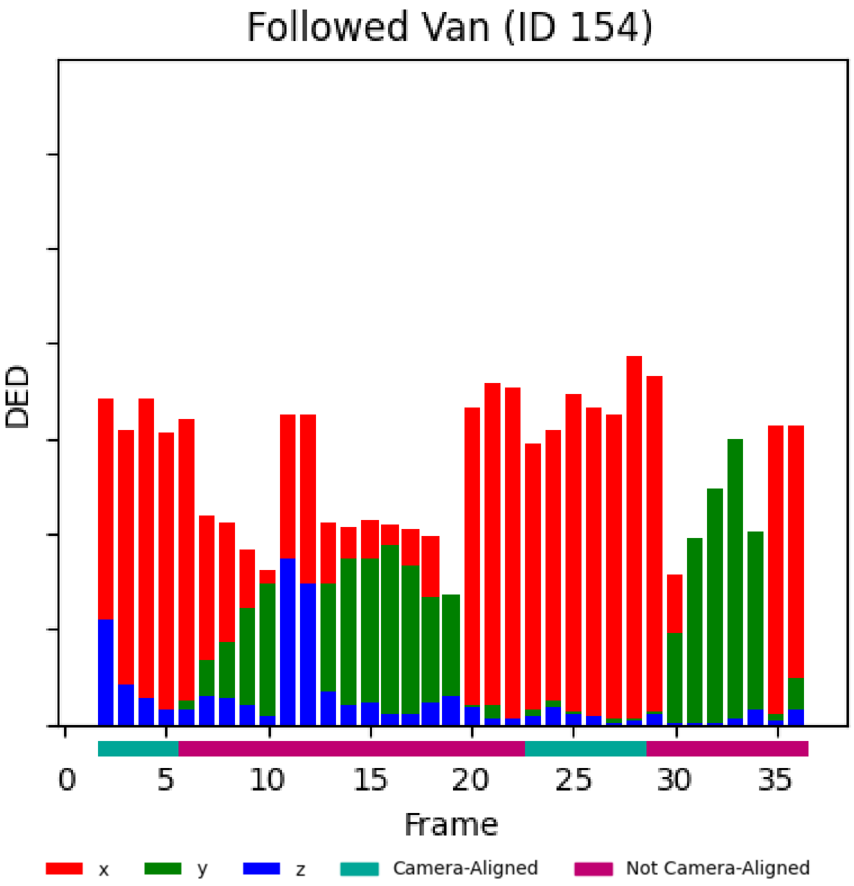
[O2] [T6]

Pipeline's Evaluation

3D Detections Evaluation

Followed Van





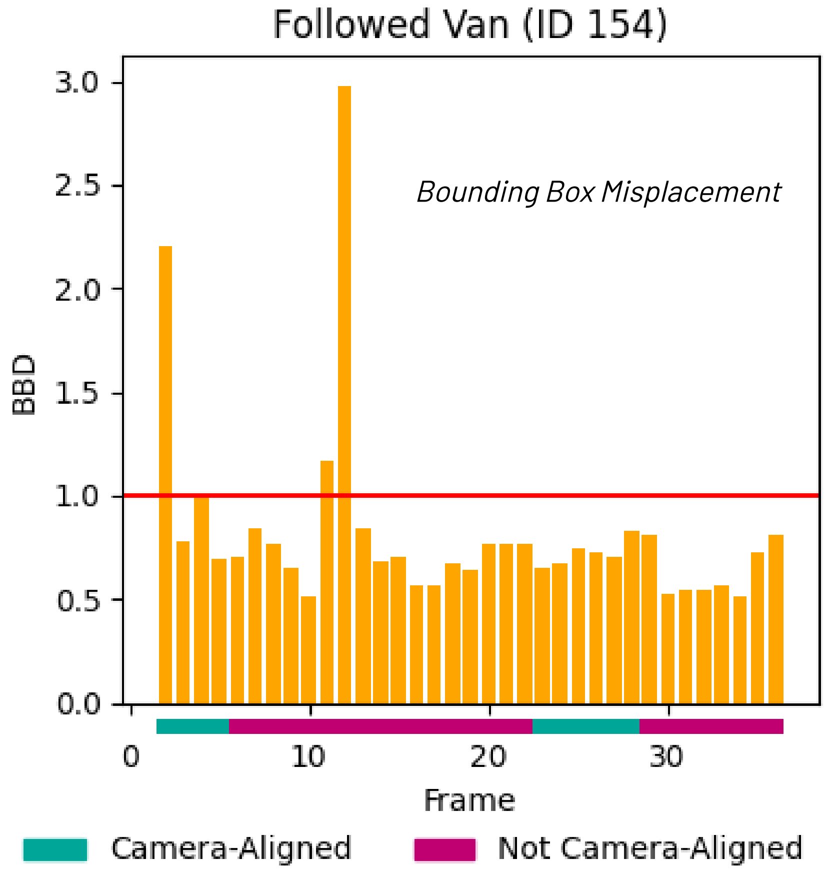
PRE-ANNOTATION PIPELINE

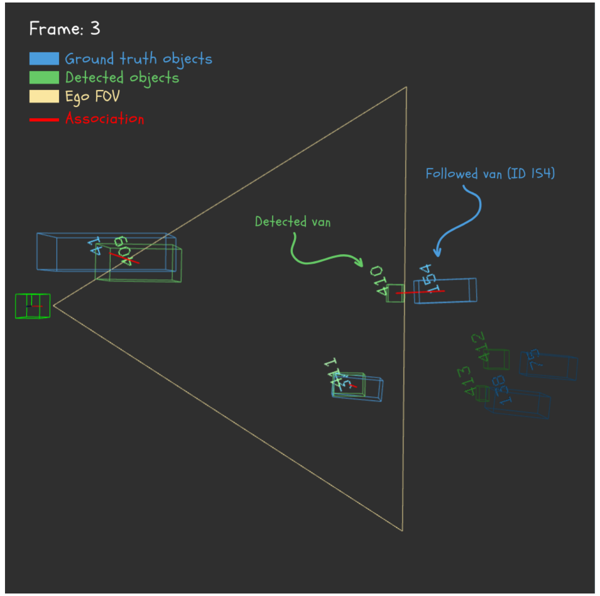
[02] [T6]

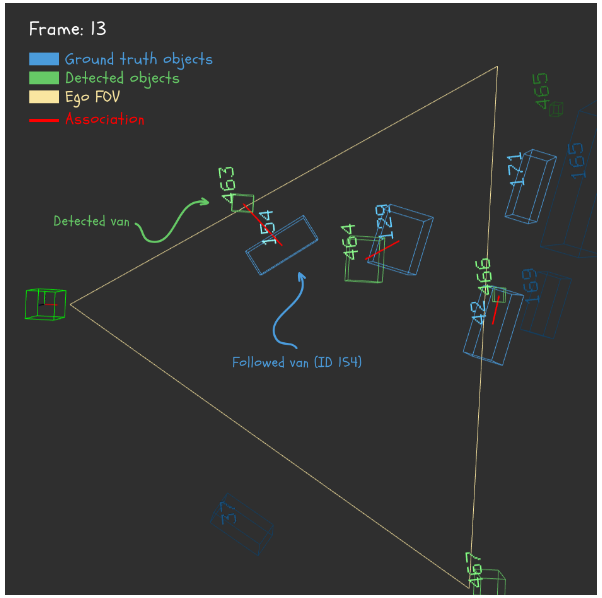
Pipeline's Evaluation

3D Detections Evaluation

Followed Van





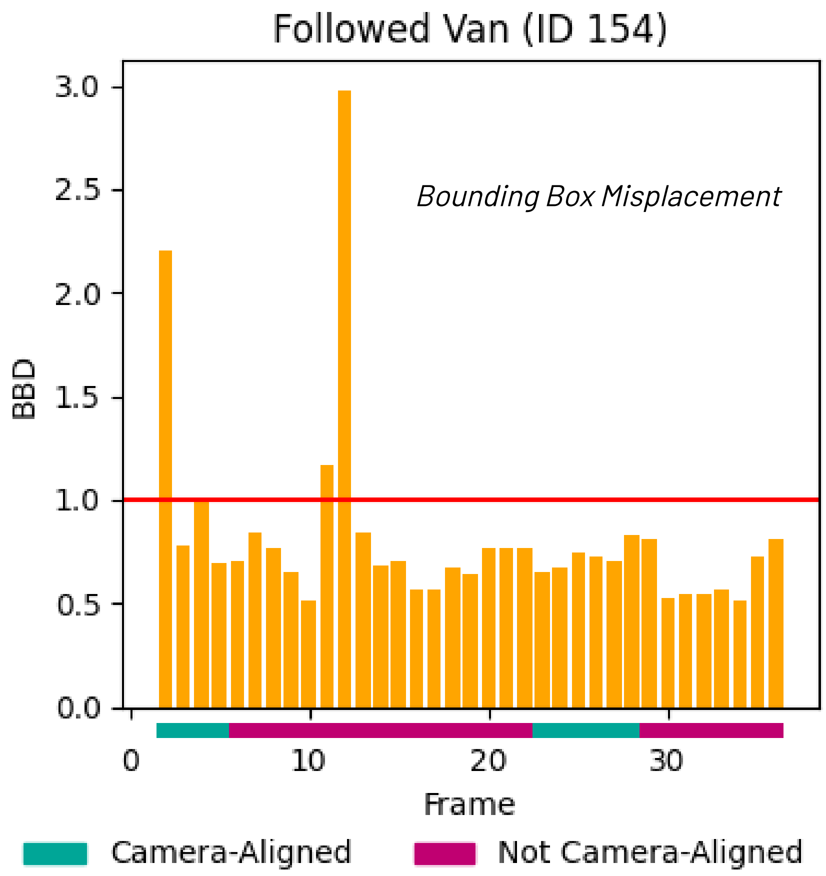


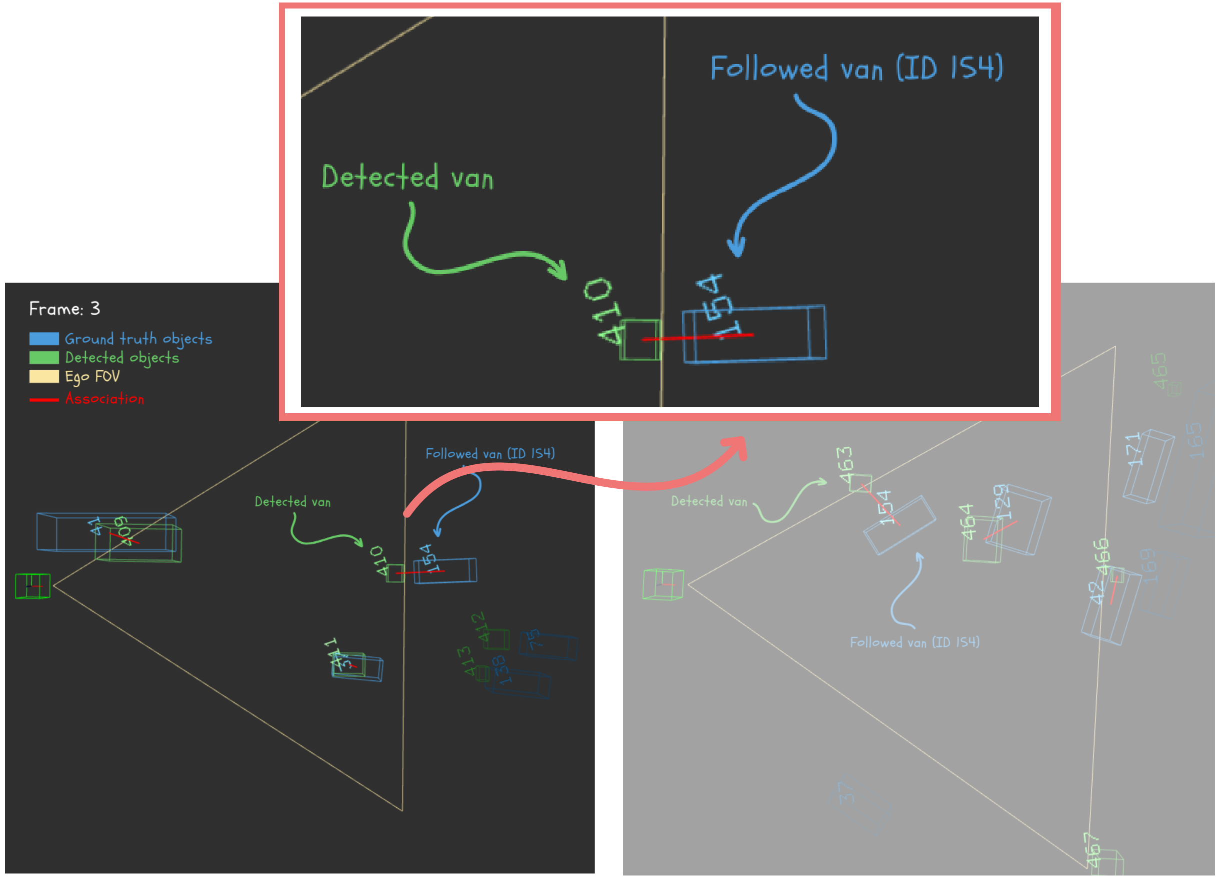
PRE-ANNOTATION PIPELINE

Pipeline's Evaluation

3D Detections Evaluation

Followed Van



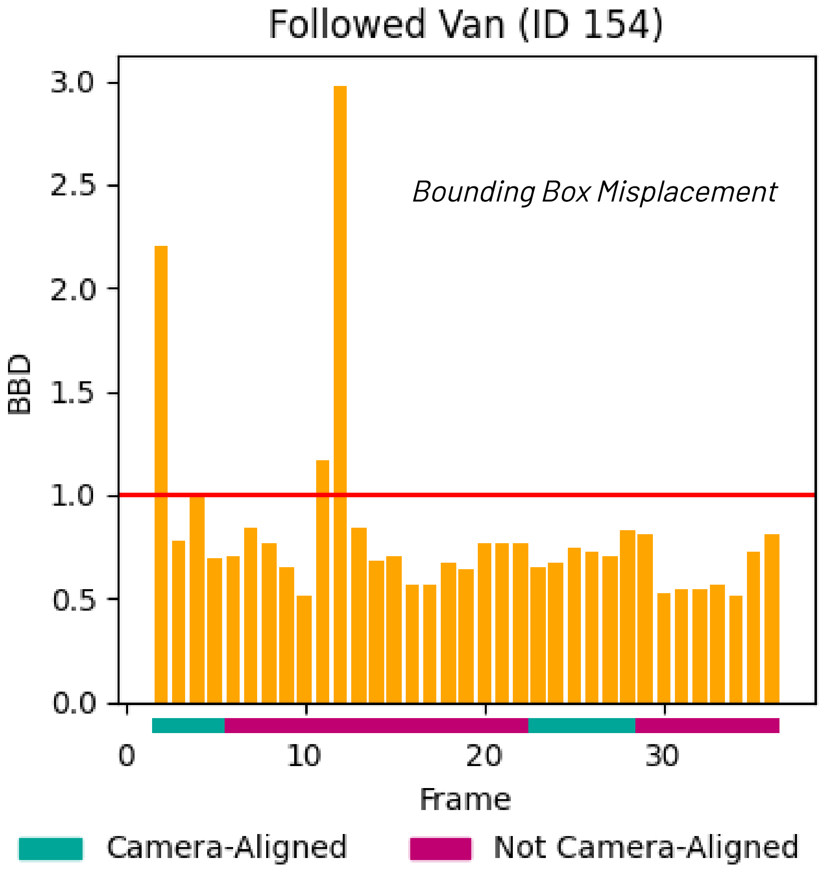


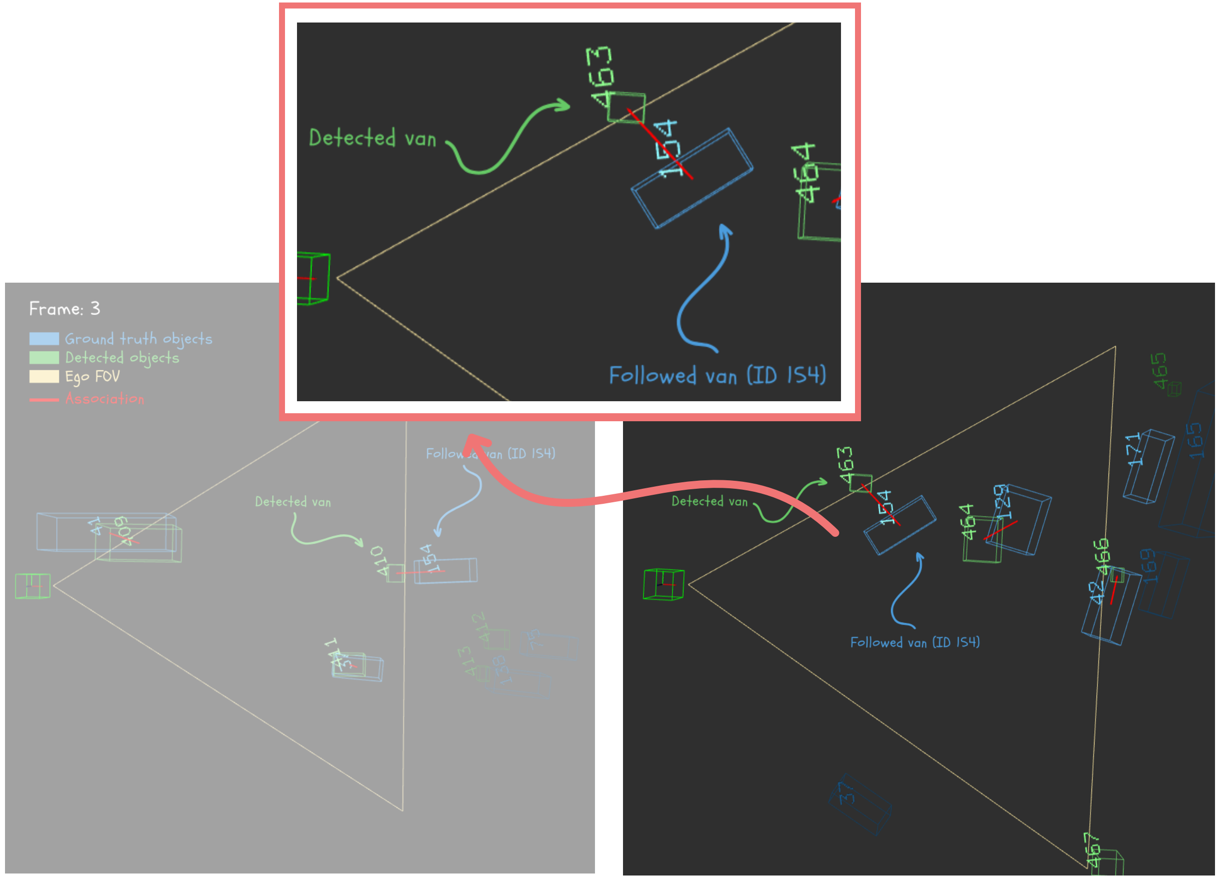
PRE-ANNOTATION PIPELINE

Pipeline's Evaluation

3D Detections Evaluation

Followed Van



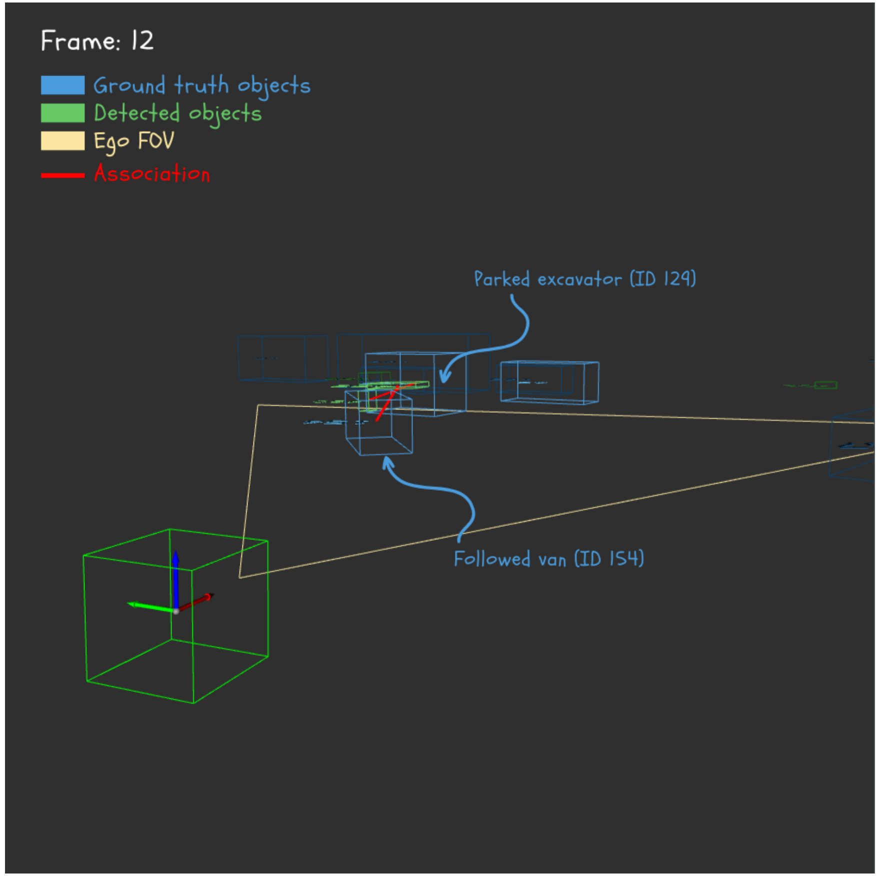


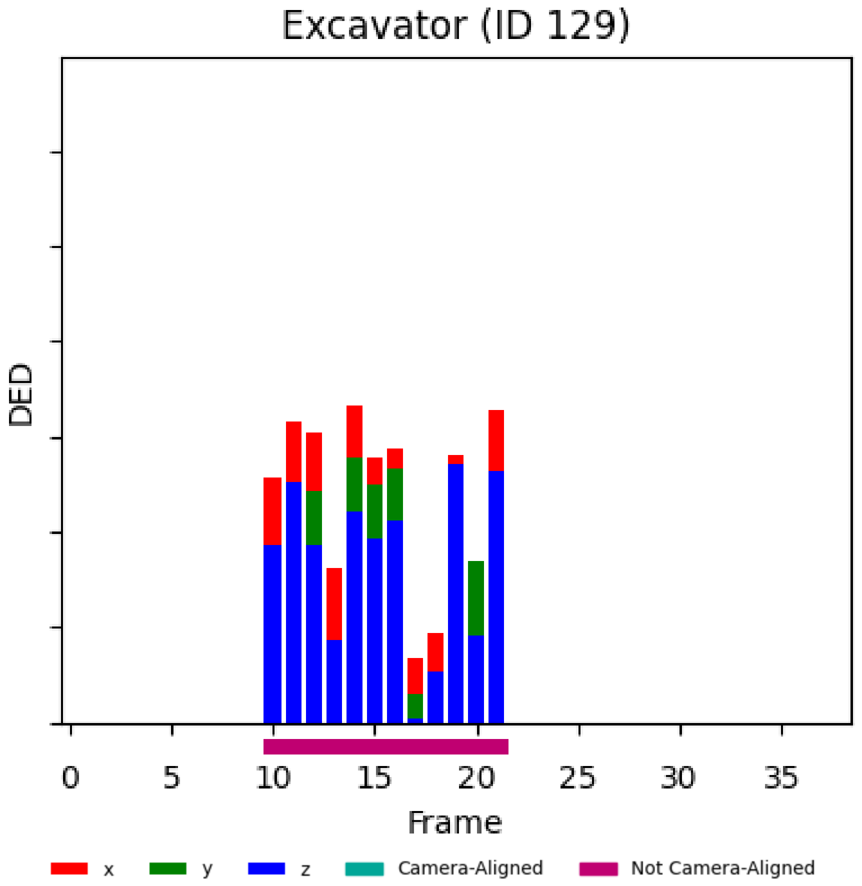
PRE-ANNOTATION PIPELINE

Pipeline's Evaluation

3D Detections Evaluation

Excavator



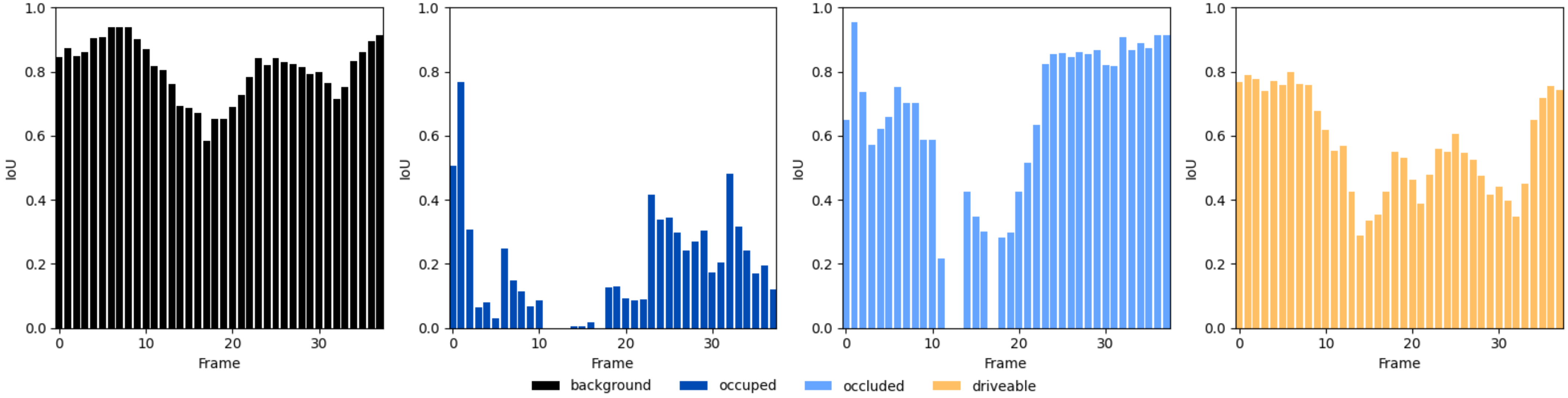


PRE-ANNOTATION PIPELINE

[O2] [T6]

Pipeline's Evaluation

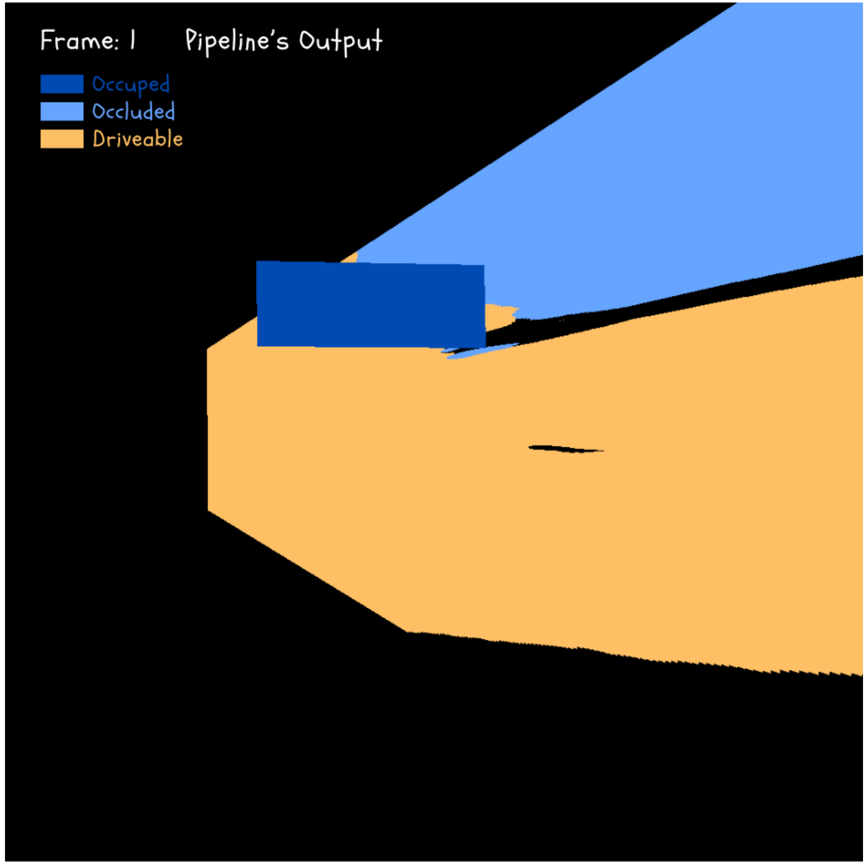
BEV Masks Evaluation

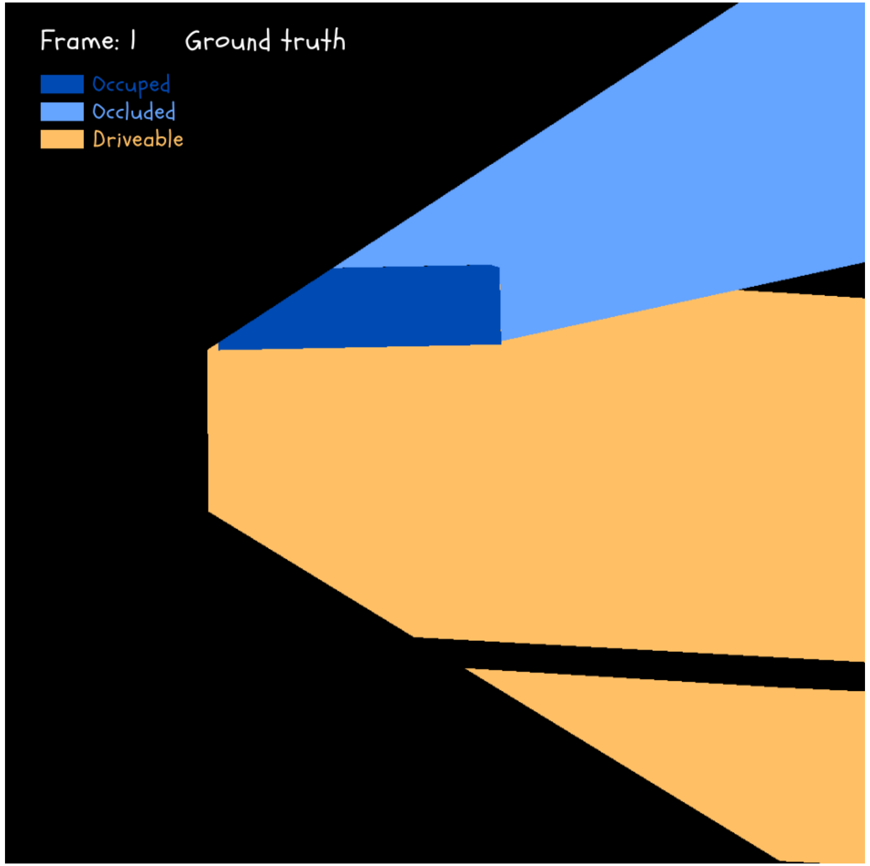


PRE-ANNOTATION PIPELINE

Pipeline's Evaluation

BEV Masks Evaluation

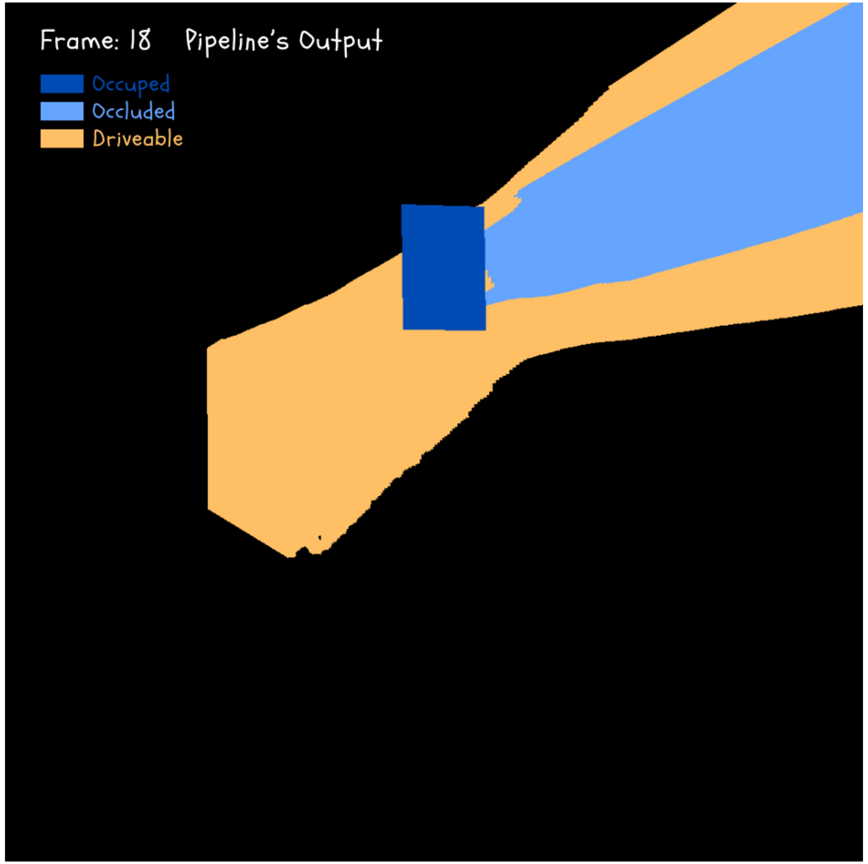


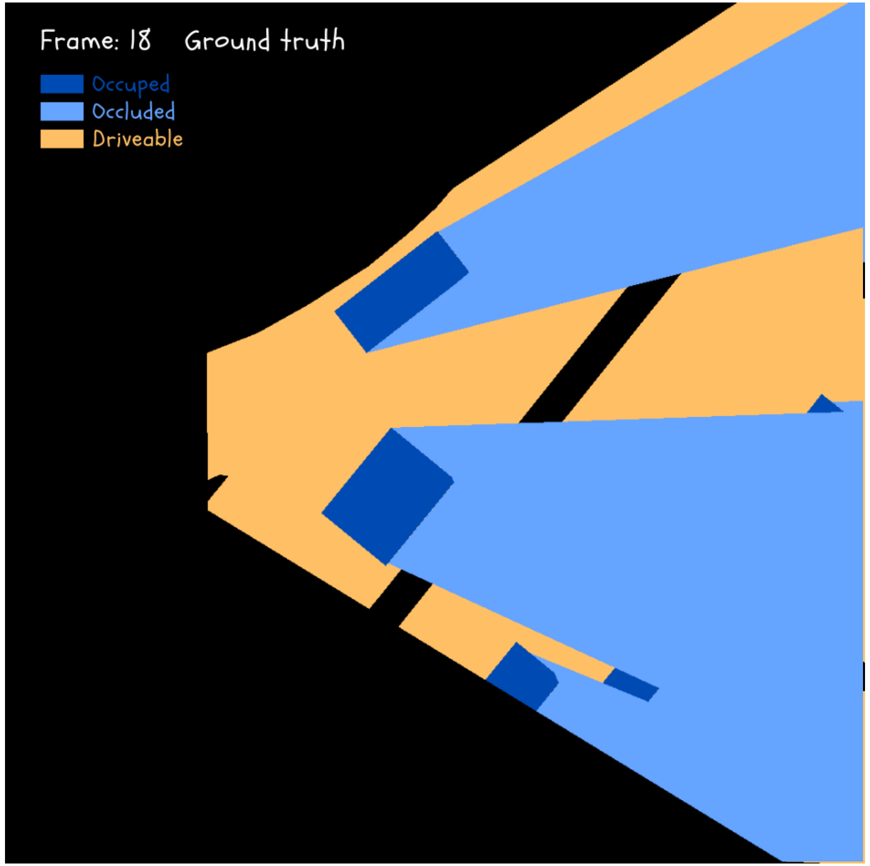


PRE-ANNOTATION PIPELINE

Pipeline's Evaluation

BEV Masks Evaluation





CONCLUSIONS

CONCLUSIONS

and Future Work

- Hypothesis on segmentation strategy not supported by evidence
- Extrinsic BEV data augmentation
- Performance and limitations of the pre-annotation pipeline

BEV2SEG_2

Analysis of Semantic Segmentation of Planar Elements in Bird's-Eye-View and Its Applications to Automated Scene Pre-Annotation

Do you have any questions?

+ DEMO